USDA
United States Department of Agriculture

Research, Education, and Economics
Agricultural Research Service

BARC, BHNRC, USNA CONFINED SPACE ENTRY PROGRAM

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BARC Director Date

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Revision Summary

Date	Revision History	Revision Class	Comment
1/10/1996	1.0	New	New Program
1/14/2015	2.0	Major	A significant change to the technical content.
10/27/2016	2.1	Minor	Clarified the meaning of the technical content.
2/14/2018	2.1	None	No changes to the meaning, language, or formatting of the technical content.
12/11/2018	2.2	Minor	Update to PRCS location at USNA. Update to CS at APU for RAS. Updates in bold

CONFINED SPACE ENTRY PROGRAM

A. SUMMARY:

This program establishes policy, responsibilities, and procedures to govern confined space entry.

B. ABBREVIATIONS:

ANSI - American National Standards Institute

ASSE - American Society of Safety Engineers

BARC – Beltsville Agricultural Research Center

BHNRC - Beltsville Human Nutrition Research Center

CFM – Cubic Feet per Minute

CFR - Code of Federal Regulations

IDLH – Immediately Dangerous to Life and Health

LEL – Lower Exclusivity Limit

NIOSH – National Institute of Safety and Health

SOHES - Safety, Occupational Health, and Environmental Staff

UEL – Upper Exclusivity Limit

USNA - United States National Arboretum

C. DEFINITION OF TERMS:

For the purposes of this program, the following definitions apply:

1. Acceptable Environmental Conditions: means the conditions that must exist in a permit space to allow entry and to ensure that employees involved with a permit-required confined space entry can safely enter into and work within the space.

2. Adequate Ventilation: Exists when the requirements of general ventilation (a change of air every three minutes) are met plus local exhaust ventilation of at least 200 cubic feet per minute (CFM) for each cutter or welder for the operation being performed in a confined space.

3. Alternative Protection Procedures: A confined space where the only hazard is atmospheric and ventilation alone can control the hazard.

4. Attendant: an individual stationed outside one or more permit spaces who monitors the authorized entrants and who performs all attendant's duties assigned in the employer's permit space program.

5. Authorized Entrant: means an employee who is authorized by the employer to enter a permit space.

6. Blanking or Blinding: The absolute closure of a pipe, line, or duct by fastening of a solid plate (such as a spectacle blind or skillet blind) that completely covers the bore and that is capable of withstanding the maximum pressure of that pipe, line, or duct with no leakage beyond the plate.

7. Confined Space:

   (a) Is large enough and so configured that an employee can bodily enter and perform assigned work.

   (b) Has limited or restricted means for entry or exit, such as tanks, vessels, silos, storage bins, hoppers, vaults, and pits.

   (c) Is not designed for continuous employee occupancy.

8. Double Block and Bleed: A method used to isolate a confined space from a line, duct or pipe by physically closing two main valves on a piping system, and opening a "vented-to-atmosphere" valve between them.

9. Emergency: Any occurrence (including any failure of hazard control or monitoring equipment) or event internal or external to the permit space that could endanger entrants.

10. Engulfment: The surrounding and effective capture of a person by a liquid or finely divided solid substance that can be aspirated to cause death by filling or plugging the respiratory system or that can exert enough force on the body to cause death by strangulation, constriction, or crushing.

11. Entrant: A person who enters a confined space to perform an assigned task.

12. Entry: The action by which a person passes through an opening into a permit-required confined space. Entry includes ensuing work activities in that space and is considered to have occurred as soon as any part of the entrant's body breaks the plane of an opening into the space.

13. Entry Permit: The written or printed document that is provided by the employer to allow and control entry into-a permit-required space and that contains specific information on that space.

14. Entry Supervisor: A person (such as the employer, foreman, or crew chief) responsible for determining if acceptable entry conditions are present at a permit space where entry is planned, for authorizing entry and overseeing entry operations, and for terminating entry as required by this section.

15. Hazardous Atmosphere: An atmosphere that may expose employees to the risk of death, incapacitation, impairment of ability to self-rescue (that is, escape unaided from a permit space), injury, or acute illness from one or more of the following causes:

    (a) A flammable gas, vapor, or mist in excess of 10 percent of its Lower Flammable Limit (LFL);

    (b) Airborne combustible dust at a concentration that meets or exceeds its LFL;

    (c) An atmospheric oxygen concentration below 19.5 percent or above 23.5 percent.

    (d) Atmospheric concentration of any substance for which a dose or a permissible exposure limit is published in Subpart G, Occupational Health and Environmental Control, or in Subpart Z, Toxic and Hazardous Substances, which could result in employee exposure in excess of its dose or permissible exposure limit;

    (e) Any atmospheric condition recognized as immediately dangerous to life or health.

16. Hazard Evaluation: A process to assess the severity of known, or real, or potential hazards of all three, at or in the confined space.

17. Hot Work: Work involving flame or spark producing operations or any work involving general heating above 205 degrees C (400 degrees F). Grinding, abrasive blasting, or other work that involve the possibility of striking sparks will be considered "Hot Work."

18. Hot Work Permit: Authorization to perform operations which could provide a source of ignition, such as riveting, welding, cutting, burning or heating.

19. Immediately Dangerous to Life or Health (IDLH): Any condition which poses an immediate threat to loss of life; may result in irreversible or immediate-severe health effects; may result in eye damage; may result in irritation or other conditions which could impair escape from the confined space.

20. Inerting: The displacement of the atmosphere in a permit-required space by a noncombustible gas (such as nitrogen) to such an extent that the resulting atmosphere is noncombustible.

21. Isolation: A process of physically interrupting and/or disconnecting pipes, lines and energy sources from the confined space.

22. Non-Permit Confined Space: A confined space that does not contain or, with respect to atmospheric hazards, have the potential to contain any hazard capable of causing death or serious physical harm.

23. Oxygen Deficient Atmosphere: An atmosphere that contains less than 19.5% oxygen by volume.

24. Oxygen Enriched Atmosphere: An atmosphere that contains more than 23.5% oxygen by volume.

25. Permit Required Confined Space: a confined space that has one or more of the following characteristics:

    (a) Contains or has a potential to contain a hazardous atmosphere;

    (b) Contains a material that has the potential for engulfing an entrant;

    (c) Has an internal configuration such that an entrant could be trapped or asphyxiated by inwardly converging walls or by a floor which slopes downward and tapers to a smaller cross-section; or

    (d) Contains any other recognized serious safety or health hazard.

26. Prohibited Condition: Any condition in a permit space that is not allowed by the permit during the period when entry is authorized.

27. Qualified Person: A person, who by reason of training, education and experience is knowledgeable in the operation to be performed and is competent to judge the hazards involved and specify controls and/or protective measures.

28. Retrieval System: The equipment (including retrieval line, chest or full body harness, wristlets, if appropriate, and a lifting device or anchor) used for non-entry rescue of persons from a permit-required space.

29. Shall**:** Denotes a mandatory requirement.

30. Should**:** A recommendation that is sound safety and health practice; it does not denote a mandatory requirement.

31. Testing**:** The process by which the hazards that may confront entrants of the permit space are identified and evaluated. Testing includes specifying the tests that are to be performed in the permit space.

D. BACKGROUND:

A Confined Space Entry Program is required by the Occupational Safety and Health Administration (OSHA). The specific program requirements are stated in 29 CFR 1910.146 and ANSI/ASSE Z117.1.

E. APPLICABILITY AND SCOPE:

The program applies to all Agriculture Research Service (ARS), Beltsville (BARC, BHNRC, and USNA) personnel that enter into confined spaces in accordance with 29 CFR 1910.146, Permit-Required Confined Spaces. The program applies to all Agriculture Research Service; Beltsville confined space entry operations. It sets forth requirements, standards, procedures, recordkeeping and reporting systems appropriate to provide safe confined space entry to Beltsville operations which require work in and around such spaces.

F. PURPOSE:

The purpose of this program is to provide specific procedures/safe work practices for employees required to enter confined spaces. These procedures/practices will be implemented in compliance with all applicable state and federal regulations pertaining to confined space entry.

G. POLICY:

No person will be permitted to enter or work in, on, or adjacent to an area that may contain or produce hazardous atmospheres except under the conditions and precautions contained in this program. Testing and Confined Space Entry Permits will not be provided for contractor operations. The contractor shall provide an Entry Supervisor and the appropriate equipment in order to test and permit the confined space in accordance with 29 CFR 1910.146, and must comply with any other applicable parts of this and other regulations listed in Paragraph O, References. The Beltsville Confined Space Entry Contractor Specification is attached as Appendix E. Where Beltsville personnel and contractor personnel are to occupy the same space for a given task or operation, the space in question will be tested and permitted for the Beltsville personnel by the affected Management Unit's Entry supervisor(s). However, such testing, permitting, and subsequent notification will in no way relieve the contractor of any pertinent statutory obligations for the safety and health of contractor personnel and the contractor will be so informed.

H. RESPONSIBILITIES:

1. Center Director(s) shall:

(a) Have ultimate responsibility for the Confined Space Entry Program for BARC, BHNRC or USNA and must provide continuing support for the Beltsville Confined Space Entry Program. 29 CFR 1910.146(c)(1) *The employer shall evaluate the workplace to determine if any spaces are permit-required confined spaces.*

(b) Ensure that the supplies and instrumentation required to perform confined space entry are budgeted for and procured. 29 CFR1910.146 (d)(4) *Provide the following equipment (specified in paragraphs (d)(4)(i) through (d)(4)(ix) at no cost to employees, maintain that equipment properly, and ensure that employees use that equipment properly.*

2. Safety, Occupational Health and Environmental Staff (SOHES) Safety Specialist shall:

(a) Provide oversight to ensure that all procedures for confined space entry are performed.

(b) Conduct random and scheduled inspections to verify the Management Units compliance with confined space entry procedures.

3. Center Directors, Research Leaders, Facilities Managers and Section Leaders will:

(a) Ensure that the supervisors under their direction, who have employees that must as part of their job function enter confined or enclosed spaces, comply with their responsibilities contained in this program.

(b) Ensure that the affected Management Unit designates personnel to serve as Entry Supervisor(s) and provide the employee(s) with the required training.

(c) Ensure that when contractors are used to perform functions in Beltsville Confined Spaces, that they comply with the requirements and are provided the information set forth in Appendix D and E. This process should be coordinated with the affected MU supervisor and Entry Supervisor(s).

(d) Ensure that instrumentation and supplies required to perform confined space testing and calibration are budgeted for and procured.

4. Supervisors shall:

(a) Request that their designated Management Unit Entry Supervisor(s) evaluate, test, issue, and post all confined spaces permits that may contain hazardous atmospheres or other hazards as defined by this program prior to allowing personnel to enter or work in, on, or adjacent to any that space.

(b) Ensure that the Management Unit Entry Supervisor(s) notify SOHES in advance of all confined space entries and that they must provide SOHES a copy of each Confined Space Entry Permit within 1 hour before the entry. Prior knowledge of confined space entry will allow SOHES to perform random spot checks to verify regulatory compliance and gather information on areas of concern in the process, so that necessary changes can be implemented if warranted.

(c) Arrange for and ensure usage of all the proper personal protective or other equipment required for work in, on, or adjacent to confined space as specified by this program. Ensure that Entry Supervisor(s), attendant(s) and authorized entrant(s) attend initial and annual certification training.

(d) Be familiar with the provisions of this program and advise all personnel under their supervision of their responsibilities.

(e) Prepare SOP(s) that address the specific entry procedures for those spaces under your control that are entered routinely.

5. Management Unit Entry Supervisor shall:

(a) Evaluate and test confined spaces within their Management Unit, issue and post confined space entry permits for those confined spaces, monitor usage of confined spaces, and perform calibration and recordkeeping procedures associated with this program.

(b) Notify the Facilities Manager and SOHES of all confined space entries, prior to entry, and provide copies of the Confined Space Entry Permit, attached as Appendix A. Utilize the Permit Required Confined Space Entry procedures attached as Appendix B.

(c) Attend the initial certification training and the annual re-certification training thereafter. Comply with the requirements of the Beltsville Respiratory Protection Program if required to use a respirator.

(d) Ensure that the instrumentation used in confined space testing is properly maintained, used, and calibrated in accordance with this program and the manufacturer's recommendations. Ensure that confined space entry equipment is available to SOHES for inspection and to download information as required. Ensure that the supervisor is kept aware of equipment funding needs, to include projected replacement parts and associated supplies needed to maintain confined space instrumentation.

(e) Ensure that, in the event that an unsafe confined space related condition is detected or suspected, work is stopped and all personnel are evacuated from the space until proper conditions and/or personal protective equipment are put into place.

(f) Notify SOHES immediately of any work stoppage, personnel evacuation, or if unable to enter due to hazardous conditions related to confined space entry and/or work being performed.

(g) Assist Management Unit supervisor in preparing Standard Operating Procedures (SOP) for entry into those spaces that are entered on a routine basis (see section H, paragraph 5, subsection (e).

6. Employees that are involved in confined space entry will:

   (a) Refrain from entering or working in, on, or adjacent to any confined or enclosed space that may contain hazardous atmospheres until it is certified SAFE by their Management Unit Entry Supervisor.

   (b) Attend the training session designed for operating and supervisory personnel which explains the hazards associated with confined spaces and the safety precautions that need to be followed.

   (c) Comply with all applicable provisions of this program and with all the instructions of their Management Unit Entry Supervisor(s), while working in, on, or adjacent to a posted confined or enclosed space.

   (d) Report to their supervisor and/or to SOHES, any condition, procedure, or equipment considered being unsafe or hazardous.

   (e) Report to their supervisor any injury, evidence of impaired health, or any condition that may affect safe performance during their assigned duties.

I. COMPLIANCE AUTHORITY:

1. The Entry Supervisors have the compliance authority to order anyone out of an area immediately or to suspend operations if they determine that an UNSAFE condition exists. Entry Supervisors will notify SOHES of this condition; in turn SOHES will promptly notify the Area Safety and Health Manager of any work stoppage and the reason for such action.

2. The Management Unit Entry supervisor(s) and SOHES are authorized access into all spaces within the confines of Beltsville for the purposes of performing official duties under the Confined Space Entry Program.

J. CONFINED SPACE ENTRY:

1. Operation Evaluation:

   Prior to starting any operation that involves entry or work on, in, or adjacent to confined spaces, the Management Unit Entry Supervisor(s) will comply with the requirements of 29 CFR 1910.146. Assistance and information is available from the SOHES Office. All operation evaluations will include, but are not limited to, the following considerations:

   (a) Contents and/or previous contents of the space that may result in the presence of oxygen-depleted or enriched atmospheres, flammable, or toxic contaminants.

   (b) Location and configuration of the space including restricted access, obstructions, remoteness, etc., which may inhibit or interfere with movement, ventilation, rescue, and firefighting efforts.

(c) Types of operations that are conducted within the space, particularly those that by the very nature of the process produce toxic and/or flammable atmospheres, oxygen depletion or enrichment, or ignition sources.

(d) Fixtures, devices, or equipment within the space that may create or contribute to hazardous conditions including piping systems, conduits, ducts, machinery, pressurized lines, etc.

(e) The presence of other hazards such as slippery surfaces, deteriorated or unstable ladders, irritant or caustic materials, etc.

(f) Boundary spaces and their contents to ensure that fire or explosion will not be caused in these spaces by the operation to be conducted.

2. Testing:

The testing and examination of a confined space will, as a minimum, incorporate the following sequentially listed procedures:

(a) Pre-calibration of all test equipment will be performed in accordance with standard operating procedures and manufacturers' recommendations. All calibrations must be logged and be signed by the calibrator.

(b) Initial testing will be performed from outside the space. Initial testing in the interior of the space may be performed by insertion of sample probes into the space.

(c) Tests for oxygen content will be conducted first utilizing an approved and calibrated oxygen meter. The oxygen level should be 20.9%, since the normal oxygen level of ambient air at sea level is 20%. Oxygen levels less than 19.5% or greater than 22% represent potentially dangerous situations and entry will not be permitted under these conditions.

(d) An approved and calibrated combustible gas indicator will be used to test for combustible vapors. The lower explosive level (LEL) for combustible vapors must be zero.

(e) Tests will be made for the presence of specific toxics dependent upon the nature of the space and its contents or previous contents. See Subpart G, Occupational Health and Environmental Control, or in Subpart Z, Toxic and Hazardous Substances for safe levels.

(f) If initial tests indicate the presence of hazardous concentrations of flammables, toxics or deviations in oxygen levels, personnel will not enter the space. The space will be ventilated and/or cleaned in a safe manner to remove flammable and toxic atmospheres and provide proper oxygen levels.

(g) If initial tests do not indicate the presence of hazardous concentrations of flammables or toxics or deviations from the normal oxygen levels, the space may be entered and the tests described in paragraphs 2 (c), (d), (e), and (f) will be performed continuously throughout the space by the Management Unit Entry Supervisor(s). The readings of the continuous monitoring (tests) will be recorded on Appendix A at fifteen (15) minute intervals. If subsequent tests indicate the presence of hazardous concentrations of flammables, toxics or deviations from normal oxygen levels; the Management Unit Entry Supervisor(s) will evacuate workers and the space will be ventilated and/or cleaned as necessary.

(h) The space will also be inspected for the presence of blisters or scale that may trap hazardous substances or for other residues that may become dangerous during the course of the work being performed. Additionally, any specific structure, equipment, or location that may be affected by the intended work will be tested and/or inspected as applicable.

(i) Post-calibration of all test equipment will be performed in accordance with standard operating procedures and manufacturers' recommendations.

3. Securing Area:

(a) All confined space entrances will be posted as a confined space and the potential hazards will be denoted, see attached Appendix A and paragraph 3 (b).

(b) All entry areas, when open, shall be guarded by a railing, pedestrian or vehicular barriers, temporary covers, or other barriers that will prevent accidental falls and foreign objects from entering the space.

4. Permitting:

(a) Upon completion of the evaluation and testing, a Confined Space Entry Permit, attached as Appendix A, will be issued by the Management Unit Entry supervisor(s). A copy of this permit will be posted at the main or most commonly used entrance to the space and at all other access areas that are open and readily accessible to personnel.

(b) A permit must be completed for all tests and distribution of copies will be done in accordance with this program. A Confined Space Entry Permit will not be in effect longer than eight (8) hours.

(c) For confined spaces where the only hazard is atmospheric and ventilation alone can control the hazard, Management Unit personnel may use "Alternative Protection Procedures" for entry. Spaces that are considered "Alternate Protection Procedure" confined spaces exempt the Management Unit personnel from using a permit. "Alternative Protection Procedure" confined spaces do require the use of the Alternative Protection Procedures attached as Appendix C.

5. Monitoring:

   Once the confined space has been certified for the operations to be performed, all work will be monitored by the Management Unit Entry Supervisor(s) to ensure that the policies, procedures, and requirements of this program are being followed.

6. Ventilating Confined Spaces:

   (a) Mechanical ventilation systems shall be set at 100% outside air. Where possible, open additional manholes, hatches, etc., to increase air circulation.

   (b) After ventilating period that provides several air changes in the confined space, repeat the testing. Continue to ventilate until testing has demonstrated that the hazardous atmosphere has be eliminated.

   (c) Ventilation air shall be clean and uncontaminated. Confined spaces will be continuously ventilated during all operations to ensure that the atmosphere within the space remains safe. Short circuiting of ventilation air flow shall always be avoided. Never ventilate a confined space where combustible gas or vapor levels have been measured to exceed 50%. When levels are above 50% LEL immediately notify the Fire and Rescue Service (9-911) and SOHES.

7. Respirators:

   (a) Respirators are required as an alternative to ventilation, or when ventilation does not adequately reduce or remove a hazardous substance for an employee, required to enter a confined space, shall be appropriate for the hazard/situation. The selection of the appropriate respirator will be made in accordance with the ARS Beltsville Respiratory Protection Program.

   (b) Supplied Air or Self-Contained Breathing Apparatus Respirators will provide:

      (1) Air capacity sufficient to perform task. This may require more than one tank depending on the task.

   NOTE: The time spent in confined space while wearing supplied air or self-contained breathing apparatus respirators must be monitored.

      (2) Reserve air supply for emergency situations.

K. STANDBY/RESCUE WORK PRACTICES:

1. An attendant will stand by at the scene of each confined space entry where a Confined Space Permit is issued and will, in case of any emergency, be able to establish immediate contact with emergency rescue personnel (9-911). This can include radio and/or telephone communication.

2. The local rescue and emergency service personnel will serve as the rescue control point for all confined space operations. Under no circumstances will unprotected, untrained persons enter a space to attempt rescue of a person overcome by toxic gases or oxygen deficiency. Rescue will be undertaken only by trained rescue personnel properly equipped to perform the task.

L. INSTRUMENTATION:

Atmospheric testing instruments require calibration and will be calibrated in accordance with manufacturer's instructions immediately prior to and after each work shift or test procedure in which the instruments are used. A log will be maintained by the Management Unit Entry Supervisor(s) and copies of all instrument calibration checks will be provided to SOHES monthly. Test instruments or their component parts when repaired or changed will be annotated in the log and on the instrument.

M. RECORDKEEPING:

1. A list of the current designated confined space areas for Beltsville is attached as Appendix F. A master Confined Space Entry Log consisting of copies of all permits issued for testing and inspections of confined spaces at Beltsville will be maintained by SOHES. The Management Unit Entry Supervisor(s) will be responsible for maintaining a log for their specific area. Permits will be maintained for a period of one (1) year from the date of the last permit issued.

2. The master copy of the records for maintenance and calibration of all confined space entry instrumentation will be kept by SOHES.

3. The master copy of the Management Unit Entry Supervisor(s) training certifications will be maintained by SOHES. Each Entry Supervisor(s) will be issued a training certification after completion of the required training and then annually after the certification training.

N. TRAINING:

1. All designated Management Unit Entry Supervisor(s) will be initially trained and certified in accordance with 29 CFR 1910.146, Permit-Required Confined Space and ANSI/ASSE Zl17.1, Safety Requirements for Confined Spaces. An annual recertification training review is required for all Entry Supervisor(s).

2. Attendants, entrants and other associated confined space personnel will only be required to receive initial training in the hazards of confined spaces and the safety precautions necessary to control such hazards.

O. REFERENCES:

1. 29 CFR 1910.146, Permit-Required Confined Space
2. 29 CFR 1910.134, Respiratory Protection
3. 29 CFR 1910.272, Grain Handling Facilities
4. 29 CFR 1910.147, The Control of Hazardous Energy (Lockout/Tagout)
5. 29 CFR 1910.Subpart Q, Welding, Cutting, and Brazing
6. ANSI/ASSE Z117.1, Safety Requirements for Entering Confined Spaces
7. COMAR 09.12.35, Confined Spaces, State of Maryland

CONFINED SPACE ENTRY APPENDICES

Appendix A – Confined Space Entry Permit and Logic Diagram

Appendix B – Permit Required Confined Space Entry Procedures

Appendix C – Alternative Protection Measures

Appendix D – Contractors

Appendix E – Contract Specifications

Appendix F – Master List of Beltsville Confined Spaces

Appendix A

Beltsville Confined Space Permit **Permit No._____**

Space to be Entered:	Purpose of Entry:
Location/Building:	Duration of Permit: (Not to Exceed 8 Hours) From: _____/_____/_____ To: _____/_____/_____

Permit Space Hazards (Check all that apply)

__ Oxygen Deficiency __ Flammable Gases/Vapors __ Airborne Dust __ Toxic Gases/Vapors
__ Mechanical Hazards __ Electrical Hazards __ Engulfment __ Other:_____

Personal Protective Equipment:
Respiratory Protection:
Rescue Equipment:
Communication Procedures:
Other:

Notification(s): (Initial when Notified)	Lockout Items:	Other Permits: (Initial if Permitted)
_____ Facilities Manager _____ SOHES _____ Rescue Service (See Below)	Electrical Y N Pipes Y N Machinery Y N	Hot Work: _____ Other:_____

Rescue Service Address: (Print Clearly)	Rescue Services Phone Numbers: Emergency: _____ Other: _____
Ventilation: (NV if Natural Ventilation Only) Capacity of Blower:_____ CFM:_____ Time Ventilation Started: _____	Atmospheric Monitor: (Monitoring Table on Back) Manufacture/Model:_____ Calibration Date: _____ Name of Calibrator:_____ (Printed)

Authorized Entrants: (Print Names)

Authorized Attendants: (Print Names)

Entry Supervisor:

I verify that all precautions have been taken and all necessary equipment has been provided for safe entry and work in this confined space. All required personnel have been notified before any work is to be started. This permit is not valid unless all appropriate items are completed. This permit to be kept at job site. Return job site copy to Safety Office following job completion.

Printed Name:_____ Date: _____/_____/_____ Time:_____

Signature:_____

******* Print All Information Neatly and Clearly*******

Appendix A

Atmospheric Monitor:

Make:_____ Model:_____ Calibration Date:_____/_____/_____

Tester's Name: _____ (Print Name) _____ (Signature)

Date :	Time:	Reason: *	Oxygen (O2) (19.5% to 23.5%)	Carbon Monoxide (CO2) < 35 ppm	Hydrogen Sulfide (H2S) < 10 ppm	Flammability Level (CH4) < 10%	Comments:
		I					
		A V					
		E					

*** Reason for Entry: I = Initial Reading; AV = After Ventilation; E = Entry into space; C = Continuous; F = Final**

**** Make Copies as Needed**

Appendix B

USDA, AGRICULTURE RESEARCH SERVICES

BARC, BHNRC, and USNA

PERMIT REQUIRED CONFINED SPACE ENTRY PROCEDURES

BEFORE ARRIVAL AT JOB SITE:

1. Entry supervisor gets permit from his/her supervisor.
2. Entry supervisor assigns jobs and reviews the jobs as a team.
3. Collects and inspects the equipment specified on the permit.
4. Turn on the air monitor and conduct the pre-calibration checks (pump check, field check with span gas, audible and visual alarm check, and then zero.)

BEFORE ENRY AT THE JOB SITE:

1. Set up traffic control and confined space equipment.
2. Zero the air monitor.
3. Perform Lockout/Tagout all energy systems.
4. Remotely monitor the air around and in the space before ventilating.
5. Ventilate as long as the Lower Explosive Level (LEL) readings are less the 50% LEL. Assume readings >50% LEL are above the Upper Explosive Level (UEL) which are too rich to burn. If the real concentration is above the UEL, ventilating the space may bring the concentration down into the flammable range (between LEL and UEL). A spark in this environment could lead to an explosion or fire. Report these readings immediately to your supervisor, SOHES, and the fire rescue services (9-911). Do not ventilate any space unless the readings are <50% LEL as indicated on the air monitoring equipment.

Appendix B (Cont.)

6. Set up and start the blower.
7. Monitor the air in the space again to ensure acceptable entry conditions.
8. When conditions are acceptable, entry supervisor signs the permit.
9. Permit is prominently posted next to the permit required confined space.
10. Check all communication equipment or review all non-verbal communications.
11. Ensure authorized entrant has donned their proper PPE is hooked up to lifeline and the retractable lifeline (if practical), steps into the opening, and communicates to the attendant as they enter the permit required confined space.
12. The Entry Supervisor(s) will continuously monitor air readings while entrant is in the confined space. These readings will be recorded on Appendix A every fifteen (15) minutes.
13. If a prohibited condition occurs, all entrants evacuate the permit confined space. The Entry Supervisor will contact SOHES of the prohibited conditions. After the space has been evacuated, steps 4 thru 8 will need to be repeated before reentry can occur.
14. When the entry operations are completed, the entry supervisor completes the permit. The permit will then be sent to SOHES for review and record keeping.

RESCUE:

1. If a prohibited condition occurs and the entrant is injured or unresponsive, the attendant will initiate rescue and emergency procedures.
2. The attendant will contact the rescue and emergency services indicated on the permit.
3. The attendant will then contact the Entry Supervisor and SOHES of the situation.
4. The attendant **WILL NOT** enter the permit required confined space to rescue the injured entrant. Only trained personnel in confined space rescue will enter the space to perform rescue actions.

Appendix C

USDA, AGRICULTURE RESEARCH SERVICES

BARC, BHNRC, and USNA

ALTERNATIVE PROTECTION PROCEDURES

For permit required confined spaces *where the only hazard is atmospheric and ventilation alone can control the hazard,* Beltsville personnel may us "alternative protection procedures" for entry. The communications manholes and the catch-basins in Beltsville fall into this category unless otherwise noted. Spaces that are considered "alternative procedure" spaces exempt the Beltsville personnel from using the Beltsville Confined Space Entry permit, a space entry team, and arrange for rescue. ARS Beltsville must ensure that workers entering such spaces have confined space entry training. To qualify for alternative procedures as par 29 CFR 1910.146(c)(5) Beltsville must:

1. Demonstrate that the only hazard of the permit space is an actual or potential hazardous atmosphere.
2. Demonstrate that ventilation alone is sufficient to maintain the permit space for entry and work to be performed within the permit required space must introduce no additional hazards.
3. Gather monitoring and inspection data to support #1 and #2.
4. If entry is necessary to conduct initial data gathering, perform such entry under the full permit required confined space program.
5. Document the determinations and supporting data and make information available to employees.

<u>ALTERNATIVE PROCEDURES CAN TAKE PLACE AFTER:</u>

a. The Entrant Supervisor(s) determines that it is safe to remove the entrance cover.
b. Guard any openings to protect against falling and falling objects.
c. Testing the atmosphere with a direct-reading calibrated air monitor the tests for oxygen levels, LEL, and potential toxic air contaminants (such as carbon monoxide, and hydrogen sulfide). Remotely monitor the air around and in the confined space before ventilating.
d. Ventilate as long as the LEL reading are less than 50%. Assume readings >50% LEL are above the UEL (too rich to burn). If the real concentration is above the UEL, ventilating the confined space may bring the concentration down into the flammable range (between LEL and UEL). Even a spark could start a fire or cause an explosion. Report these reading to immediately to your

Appendix C (Cont.)

supervisor, SOHES, and rescue service. Do not ventilate unless the readings are >50% LEL. With LEL readings that high, such a space would not be classified as an "alternate procedure" space.

e. Set up and run the blower.

f. Monitor the air in the confined space again to ensure acceptable entry conditions.

g. Continuously ventilate and monitor the confined space to ensure the air remains without hazards whenever any Beltsville personnel enter the confined space.

h. Employees must exit immediately if a hazardous atmosphere is detected during entry, and the confined space must be evaluated to determine how the hazardous atmosphere developed.

i. While OSHA's alternate procedures require only one person for entry, these are the minimum requirements. Alternative procedures at Beltsville are conducted with two personnel, an entrant and an attendant. There must be an emergency action and rescue plane in effect prior to entry.

Appendix D

CONTRACTORS

OSHA also has requirements whenever contractors are performing Permit Required Confined Space (PRCS) work your site. These requirements can be found in 1910.146(c)(8) and include:

1. Inform the contractor that the workplace contains PRCSs and the entry is only allowed through compliance with a PRCS program that meets the following requirement:
    a. The employer must inform the contractor of the elements, including the hazards identified and the host employer's experience with the space and why the space is a PRCS.
    b. Inform the contractor of any precautions or procedures that the host employer has implemented for the protection of workers in or near PRCSs where contractor personnel will be working.
    c. Coordinate entry operations with the contractor when both host employer personnel and contractor personnel will be working in or near PRCS as per 1910.146(d)(11): Develop and implement procedures to coordinate entry operations when employees of more than one employer are working simultaneously as entrants in the PRCS so that employees of one employer do not endanger the employees of any other employer.
    d. Debrief the contractor at the conclusion of the entry operations regarding the PRCS program followed and regarding any hazards confronted or created in the PRCS during entry operations.

In addition to complying with PRCS requirements that apply to all employers, easch contractor as per 29 CFR 1910.146(c)(9) must:

1. Obtain any available information regarding PRCS hazards and entry operations from the host employer.
2. Coordinate entry operations with the host employer when both host employer personnel and contractor personnel will be working in or near PRCS as per 1910.146(d)(11): Develop and implement procedures to coordinate entry operations when employees of more than one employer are working simultaneously as entrants in the PRCS so that employees of one employer do not endanger the employees of any other employer.
3. Inform the host employer of the PRCS program that the contractor will follow and of any hazards confronted or created in permit spaces, either through a debriefing or during the entry operation.

Appendix E

CONFINED SPACE ENTRY CONTRACT SPECIFICATIONS

INTRODUCTION: Applicability

0.1.1 The Federal Department of Labor Occupational Safety and Health Administration (OSHA) confined space entry regulations (29 CFR 1910.147) are attached to this specification and shall be considered part of this specification.

0.1.2 These requirements apply to all business entities contracted by USDA, Agricultural Research Service, Beltsville Agricultural Research Center (BARC) (referred to in this document as the owner or owner's representative), engaged in any confined space entry projects.

0.1.3 The owner or owners' representative reserves the right to monitor all health and safety aspects of the project and to halt such projects at any time the work in progress poses an undue risk to the workers health or the buildings occupants, or the specifications are not being complied with or that damage to the owner's property may occur or has already occurred.

0.1.4 If found in non-compliance with these specifications, the contractor or the contractors' representative on site will be given verbal notice with written confirmation. The written notification will serve as a warning to the contractor.

0.1.5 In the event that the work is stopped due to any reasons noted in 0.1.3, such discontinuance of work shall not constitute a valid reason for failure to comply with the work schedule and the agreed completion date, or cause any additional costs to be incurred by the owner.

0.1.6 If work is stopped, the work will not resume until all deficiencies are corrected to the satisfaction of the owner/owner representative.

0.1.7 Repeated non-compliance with the specifications requirements and/or shutdown of the work related to such non-compliance will be cause for termination of the contract.

0.1.8 The owners' inspection procedures in no way relieves the contractor of any responsibilities as defined in this specification, or in the applicable OSHA regulations.

0.1.9 The contractor will complete all sections of the Beltsville confined space permit prior to entry into the confined space. This Beltsville permit will serve as written permission for the contractor to enter the confined space and must be signed by the contractor's designated Entry Supervisor.

Appendix E (Cont.)

0.2 Deviations

0.2.1 Any deviation from these specifications must be approved in writing by the owner or representatives of the owner prior to issuance of the contract. Any request for deviation must be submitted in writing to the owner/owner's representative.

0.2.2 If a deviation of contractual agreement is necessary once work has commenced, it must be approved by the owner/owner's represent prior to the implementation of such change.

0.2.3 Any deviations to the contract or specifications requested by the contractor and accepted by the owner or their representatives, the contractor shall be responsible for all affected work and costs.

PART 1 – GENERAL

1.1.0 SCOPE. This specification covers any work which involves the entry of workers into confined spaces which include but are not limited to underground tanks and vaults, sewer systems, steam manholes, storm drain systems, underground electrical and signal cable distribution, and any other spaces that met the OSHA definition of a confined space as per OSHA 1910.146.

1.1.1 The requirements of the General Conditions and outlier documents comprising the Contract apply to work of this section.

1.1.2 Penalties or fines from regulatory agencies against Beltsville due to the contractors' non-compliance with federal, state or local regulations, the contractor will reimburse Beltsville for all penalties and fines and other related expenses.

1.2.1 DESCRIPTION OF WORK

1.2.2 Describe the scope of work, who will do the work, how the tasks will be done, where will the work be done, and when will the work be completed.

PART 2 – SUBMITTALS AND NOTICES

2.1.0 Prior to Commencement of Work the contractor shall submit the following documents to the USDA Beltsville Safety and Health Officer:

2.1.1 Submit the contractor's written confined space entry program as per 29 CFR 1910.146(c)(4).

Appendix E (Cont.)

2.1.2 Submit documentation in the form of certificates containing the employee's name, signature of trainers, and dates of training to the Owner indicating that all employees have received training on confined spaces as per OSHA section 1910.146(g).

2.1.3 In addition, submit documentation that the firm, employees, and subcontractors are familiar with the current regulations of the U.S. Department of Labor, Occupational Safety and Health Administration (OSHA)

2.1.4 In the advent that respiratory protection shall be used; submit individual certificates, signed by a M.D., for each employee, indicating that said employee is in proper medical condition to use a respirator. Submit individual documentation which type and model of respirator an employee can wear. Submit a copy of the contractor's written respiratory protection program and evidence of employee training as per OSHA 1910.134

2.1.5 Submit a copy of the contractors written Hazard Communication Program and Safety Data Sheets (SDS) for all materials to be used by the contractor on site.

PART 3 – MATERIALS AND EQUIPMENT

3.1.0 The contractor must provide all equipment and materials necessary to meet the terms of this specification and the OSHA confined space regulations.

3.1.1 Deliver all materials in the original packages, containers, or bundles bearing the name of the manufacturer and the brand name.

3.1.2 Store all materials subject to damage off the ground, away from wet or damp surfaces, and under cover sufficient to prevent damage or contamination.

3.1.3 Damaged or deteriorating materials shall not be used and shall be removed from the premises.

3.1.4 Provide and maintain a sufficient quantity of all types of materials, tools and equipment for confined space entry including but not limited to the following:

1. A Fall Protection and Retrieval system that is certified by the manufacturer as man-rated and all parts of the system must withstand 5000 pounds of force. The system will include at a minimum a winch, a separate retractable lifeline, and full body harnesses worn by each worker. The contractor must certify that the winch system has not been utilized nor will it be used for the movement of any objects, equipment, or machinery.

Appendix E (Cont.)

2. A confined space blower to be used to ventilate all confined spaces prior to entry by providing positive pressure ventilation.

3. An escape self-contained breathing apparatus provided to each entrant when the potential for a hazardous atmosphere exists.

4. Entrants and attendants will be provided with hard hats, protective gloves, protective clothing, and steel toe boots. The contractor will provide the necessary protective equipment to adequately address the hazard of each confined space.

5. An air monitor that is capable of simultaneously measuring percent LEL, carbon monoxide, hydrogen sulfide, and oxygen in the range of concentrations addressed by this specification and the OSHA regulations. The monitor will have the necessary attachments to provide remote confined space testing. The contractor will maintain and calibrate the monitor as per the manufacturer's recommendations.

6. Intrinsically safe flashlights and other lighting equipment.

7. Communication equipment for notifying the rescue services and for communication between the attendant and the entrants.

PART 4 – ENTRY PERMITS

4.1 The contractor will conduct all confined space entry operations in accordance with the USDA Beltsville confined space entry permit system. A copy of the USDA Beltsville confined space permit is attached to this specification.

4.1.1 The entry permit shall be completed and approved by the contractor's entry supervisor prior to employee entry into the confined space.

4.1.2 A permit will not be valid for a period of more than 8 hours or one session of entry into a confined space. After a permit has expired a new permit will be issued.

4.1.3 During the confined space entry, the entry permit will be displayed immediately adjacent to the confined space. A copy of the permit will be provided to the USDA Beltsville Safety Office.

4.1.4 The requirements of the entry permit must be fulfilled prior to entry into the space.

Appendix E (Cont.)

PART 5 – HOT WORK PERMIT

5.1 Hot work is highly hazardous and is not permitted inside a confined space except by special written permission by the USDA Beltsville Facilities Services via an "open flame permit".

PART 6 – LOCKOUT/TAGOUT

6.1.0 Before entry into the confined space, the space must be isolated lines and pipes which may introduce hazards. Isolation must be achieved by closing off or disconnecting by means of closing valves and locking and tagging, or by blanking and tagging as per OSHA section 1910.147.

6.1.1 Mechanical equipment or apparatus in the confined space which may create a hazard shall be disconnected or disabled, and blocked and locked to prevent movement.

6.1.2 Electrical equipment or wiring which may create a hazard shall be de-energized, locked and tagged.

6.1.3 Devices such as padlocks shall be provided and used for lockout and isolation. The only keys to the locks shall be in the possession of the persons entering the confined space.

6.1.4 Prior to starting work the contractor will verify that de-energizing or isolation has been accomplished.

6.1.5 At the completion of the project the contractor will ensure that all isolated systems are returned to service.

PART 7 – VENTILATING CONFINED SPACES

7.1.0 Mechanical ventilation systems shall be set at 100% outside air. Where possible, open additional manholes to increase air circulation.

7.1.1 After a ventilating period that provides several air changes in the confined space, repeat the testing. Entry may not begin until testing has demonstrated that the hazardous atmosphere has been eliminated.

7.1.2 Ventilation air shall be clean and uncontaminated Oxygen or compressed air shall never be used for ventilating a confined space.

7.1.3 Short circuiting of ventilation air flow shall be always be avoided.

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7.1.4 Never ventilate a confined space where combustible gas or vapor levels have been measured to exceed 50%. When levels are above 50 % LEL immediately notify the Rescue Service and the BA Safety office.

PART 8 – AIR SAMPLING MONITORING

8.1.0 All air monitoring methods and equipment used by the contractor will be approved by an Industrial Hygienist Certified by the American Board of Industrial Hygiene. Testing will be conducted in accordance with appendix B of the OSHA standard 1910.146. A copy of this appendix is attached to this specification and is part of this specification.

8.1.1 Throughout the entire project for each confined space entry the Contractor shall continuously conduct air monitoring for percent lower explosive limit (%LEL), hydrogen sulfide (H2S), carbon monoxide (CO), and oxygen. The measurements will be entered on the USDA Beltsville confined space permit posted at the entry location at fifteen (15) minute intervals.

8.1.2 Air testing will be conducted in the following sequence:

1. Prior to entering the tank using remote sampling method.
2. After ventilating the tank for sufficient time to provide several air changes in the vault.
3. During worker entry of the tank
4. Continuously while workers are in the tank.

8.1.3 Air testing shall be conducted by a person trained and experienced in the use of the confined space entry air monitoring equipment. The person performing the tests will record all test results and test times on the posted USDA Beltsville confined space permit and will initial the appropriate section of the permit.

8.1.4 The air monitoring equipment will be certified as intrinsically safe by the authority having jurisdiction and will be used according to the manufacturer’s recommendations.

8.1.5 Prior to testing the air the contractor will perform the following tests at least daily:

1. Field calibration of the monitor for LEL, oxygen, hydrogen sulfide, and carbon monoxide.
2. Zero adjustment of the monitor for Oxygen, LEL, hydrogen sulfide, and carbon monoxide in clean air.

3. Field testing of the monitor alarms to verify an audible and visual warning when contaminant levels are at or above 10 % LEL, 10 parts per million (ppm) of hydrogen sulfide, 35 ppm of carbon monoxide and when oxygen levels are outside the range of 19.5 to 23.5 percent.
4. Testing of the monitor's remote air pump system and related tubing to verify its integrity.

8.1.6 The contractor will also be responsible for monitoring airborne combustible dust levels in the confined space whenever conditions or work activity expected to generate airborne dust.

8.1.7 The contractor will monitor the concentrations of any atmospheric contaminants for which a dose or a permissible exposure limit is published in Subpart G, Occupational Health and Environment Control, or in Subpart Z, Toxic and Hazardous Substances, of the OSHA regulations for which a threshold limit value is published in the most recent edition of the American Conference of Governmental Industrial Hygienists (ACGIH) threshold limit value (TLV) booklet.

8.1.8 If at any time during the confined space work air monitoring levels >10 % LEL, 10 ppm hydrogen sulfide, 35 ppm carbon monoxide, or if oxygen levels are outside the range of 19.5% to 23.5% all workers will immediately evacuate the confined space and work will discontinue until the Beltsville Safety Office has been notified and appropriate steps have been taken to bring the space into acceptable levels.

NOTE: In the event work is discontinued due to unacceptable contaminants levels such discontinuance shall not constitute valid reason for failure to comply with work schedule, cost estimates, and completion date.

8.1.9 The monitoring of each confined space may also be done by the Owner or their representative. This shall in no way obviate the Contractor's responsibilities to conduct monitoring under the law, this specification, and/or the OSHA regulations.

GLOSSARY OF TERMS

1. Acceptable entry conditions - means the conditions that must exist in a permit space to allow entry and to ensure that employees involved with a permit-required confined space entry can safely enter into and work within the space.

2. Attendant - means an individual stationed outside one or more permit spaces who monitors the authorized entrants and who performs all attendant's duties assigned in the employer's permit space program.

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3. Authorized entrant - means an employee who is authorized by the employer to enter a permit space.

4. Blanking or blinding - means the absolute closure of a pipe, line, or duct by the fastening of a solid plate (such as a spectacle blind or a skillet blind) that completely covers the bore and that is capable of withstanding the maximum pressure of the pipe, line, or duct with no leakage beyond the plate.

5. Confined space - means a space that:

(a) Is large enough and so configured that an employee can bodily enter and perform assigned work;

(b) Has limited or restricted means for entry or exit (for example, tanks, vessels, silos, storage bins, hoppers, vaults, and pits are spaces that may have limited means of entry.);

(c) Is not designed for continuous employee occupancy.

6. Double block and bleed - means the closure of a line, duct, or pipe by closing and locking or tagging two in-line valves and by opening and locking or tagging a drain or vent valve in the line between the two closed valves.

7. Emergency - means any occurrence (including any failure of hazard control or monitoring equipment) or event internal or external to the permit space that could endanger entrants.

8. Engulfment - means the surrounding and effective capture of a person by a liquid or finely divided (flowable) solid substance that can be aspirated to cause death by filling or plugging the respiratory system or that can exert enough force on the body to cause death by strangulation, constriction, or crushing.

9. Entry - means the action by which a person passes through an opening into a permit-required confined space. Entry includes ensuing work activities in that space and is considered to have occurred as soon as any part of the entrant's body breaks, the plane of an opening into the space.

10. Entry permit - means the written or printed document that is provided by the Beltsville Area to allow and control entry into a permit space and that contains the information specified in paragraph (f) of 1910.146.

Appendix E (Cont.)

11. Entry supervisor - means the person (such as the contractor, employer, foreman, or crew chief) responsible for determining if acceptable entry conditions are present at a permit space where entry is planned, for authorizing entry and overseeing entry operations, and for terminating entry as required by this section. The entry supervisor must be an employee of the contractor.

> **NOTE**: An entry supervisor also may serve as an attendant or as an authorized entrant, as long as that person is trained and equipped as required by this section for each role he or she fills. Also, the duties of entry supervisor may be passed from one individual to another during the course of an entry operation.

12. Fall Protection - means a retractable lifeline, retrieval winch, and harnesses used to protect workers from a fall and to get entrants out of the confined space during an emergency.

13. Hazardous atmosphere - means an atmosphere that may expose employees to the risk of death, incapacitation, and impairment of ability to self-rescue (that is, escape unaided from a permit space), injury, or acute illness from one or more of the following causes:

(a) Flammable gas, vapor, or mist in excess of 10 percent of its lower flammable limit (LFL);

(b) Airborne combustible dust at a concentration that meets or exceeds its LFL;

> **NOTE**: This concentration may be approximated as a condition in which the dust obscures vision at a distance of 5 feet (1.52 m) or less.

(c) Atmospheric oxygen concentration below 19.5 percent or above 23.5 percent;

(d) Atmospheric concentration of any substance for which a dose or a permissible exposure limit is published in Subpart G, Occupational Health and Environmental Control, or in Subpart Z, Toxic and Hazardous Substances, of this Part and which could result in employee exposure in excess of its dose or permissible exposure limit;

> **NOTE**: An atmospheric concentration of any substance that is not capable of causing death, incapacitation, impairment of ability to self-rescue, injury, or acute illness due to its health effects is not covered by this provision.

(e) Any other atmospheric condition that is immediately dangerous to life or health.

Appendix E (Cont.)

NOTE: For air contaminants for which OSHA has not determined a dose or permissible exposure limit, other sources of information, such as Material Safety Data Sheets that comply with the Hazard Communication Standard, section 1910.1200 of this Part, published information, and internal documents can provide guidance in establishing acceptable atmospheric conditions.

14. Hot work permit - means the employer's written authorization to perform operations (for example, riveting, welding, cutting, burning, and heating) capable of providing a source of ignition.
Immediately dangerous to life or health (IDLH) - means any condition that poses an immediate or delayed threat to life or that would cause irreversible adverse health effects or that would interfere with an individual's ability to escape unaided from a permit space.

NOTE: Some materials — hydrogen fluoride gas and cadmium vapor, for example — may produce immediate transient effects that, even if severe, may pass without medical attention, but are followed by sudden, possibly fatal collapse 12-72 hours after exposure. The victim "feels normal" from recovery from transient effects until collapse. Such materials in hazardous quantities are considered to be "immediately" dangerous to life or health.

15. Inerting - means the displacement of the atmosphere in a permit space by a noncombustible gas (such as nitrogen) to such an extent that the resulting atmosphere is noncombustible.

NOTE: This procedure produces an IDLH oxygen-deficient atmosphere.

16. Isolation - means the process by which a permit space is removed from service and completely protected against the release of energy and material into the space by such means as: blanking or blinding; misaligning or removing sections of lines, pipes, or ducts; a double block and bleed system; lockout or tagout of all sources of energy; or blocking or disconnecting all mechanical linkages.

17. LEL (Lower Explosive Limit) - the minimum concentration of a combustible gas or vapor in air which will ignite if an ignition source is present.

18. Line breaking - means the intentional opening of a pipe, line, or duct that is or has been carrying flammable, corrosive, or toxic material, an inert gas, or any fluid at a volume, pressure, or temperature capable of causing injury.

19. Lockout/Tagout - The mechanical disconnecting, securing, and labeling, with warning tags, of mechanical, electrical, or pressurized equipment to reduce the likelihbp^ of inadvertent activation or the release of dangerous energy.

Appendix E (Cont.)

20. Monitoring and testing - means the process by which the hazards that may confront entrants of a permit space are identified and evaluated. Testing includes specifying the tests that are to be performed in the permit space.

> **NOTE**: Testing enables employers both to devise and implement adequate control measures for the protection of authorized entrants and to determine if acceptable entry conditions are present immediately prior to, and during, entry.

21. Oxygen Deficient Atmosphere - means an atmosphere containing less than 19.5 percent oxygen by volume.

22. Oxygen Enriched Atmosphere - means an atmosphere containing more than 23.5 percent oxygen by volume.

23. Permit-Required Confined Space - means all confined spaces that require a permit to enter.

24. Permit-Required Confined Space Program - means the contractor's/employer's overall program for controlling, and, where appropriate, for protecting employees from, permit space hazards and for regulating employee entry into permit spaces.

25. Positive Pressure Ventilation - The use of air forced under pressure to provide and circulate fresh repairable air into the area, simultaneously removing stale or contaminated air.

26. Prohibited condition - means any condition in a permit space that is not allowed by the permit during the period when entry is authorized.

27. Rescue service - means the personnel designated to rescue employees from confined spaces. At the Beltsville Agricultural Research Center (BARC), rescue services will be provided by the Prince George's County Fire & Rescue.

28. Respirator - A device designed to protect the wearer from inhaling harmful atmospheres either by purifying contaminated air using filters or adsorbent cartridges or by supplying respiratable air.

29. Retrieval system - means the equipment (including a retrieval line, chest or full-body harness, wristlets, if appropriate, and a lifting device or anchor) used for non-entry rescue of persons from permit spaces.

Appendix F

MASTER LIST OF BARC, BHNRC, and USNA CONFINED SPACES

Note: This list is presented, as best and accurately as it can, by the group(s) responsible for the spaces. This does not prohibit or otherwise preclude entry of these spaces by others but just assigns them to group(s) most likely to be required to enter.

Contents:

1. Facilities Services, Physical Plant Management

2. BARC, Research Support Services

3. BARC, Information Technology

4. USNA, Facilities Unit

Appendix F

Facilities Services

Physical Plant Management

USDA/BARC PRCS INVENTORY
WEST SIDE/AREA B
ELECTRIC POWER CABLE RUN

Table 1

PRCS # West/Side/B Electric Power (WBEP)	Location
1	South of Bldg. 014; 36" cast iron lid (Not in service)
2	South of Bldg. 014; 30" cast iron lid (Not in service)
3	Not in service
4	Northeast of new substation
5	East of new substation
6	East of new substation
7	Southeast of new substation (Not in service)
8	Northwest corner of Bldg. 050 (Not in service)
8A	North of Bldg. 050, HH 2 (Not in service)
9	Northeast corner of Bldg. 050 (Not in service)
10	Not in service
11	Not in service
12	Not in service
13	Not in service
14	Northeast corner of Bldg. 009 (Not in service)
15	Not in service

USDA/BARC PRCS INVENTORY
WEST SIDE/AREA B
ELECTRIC POWER CABLE RUN

Table 1

PRCS # West/Side/B Electric Power (WBEP)	Location
16	North of Bldg. 010A
17	East of Bldg. 011, HH9
18	Southeast of Bldg. 010A
19	Southwest of Bldg. 006
20	North parking lot Bldg. 007
21	South parking lot Bldg. 007
21A	North of Bldg. 007
22	Southeast corner of Bldg. 006
23	Southwest corner of Bldg. 002
24	Southeast corner of Bldg. 002
25	Southeast corner of Bldg. 002
26	East of Bldg. 002
26A	East of Bldg. 002
27	East of Bldg. 003
28	East of Bldg. 004

USDA/BARC PRCS INVENTORY
WEST SIDE/AREA B
ELECTRIC POWER CABLE RUN

Table 1

PRCS # West/Side/B Electric Power (WBEP)	Location
29	South of Bldg. 005
30	Southeast corner of Bldg. 005
31	East of Bldg. 002, Main front lawn
32	Northeast of Bldg. 002, Main front lawn (Not in service)
33	Not in service
34	North of Bldg. 009, HH4 (Not in service)
35	Not in service
36	Southeast of Bldg. 010A
37	Southwest of Bldg. 006 (Not in service)
38	Southeast of Bldg. 006 (Not in service)
39	Not in service
40	Not in service
41	Southeast of Bldg. 002
42	East of Bldg. 010
42A	North of Bldg. 010A

USDA/BARC PRCS INVENTORY
WEST SIDE/AREA B
ELECTRIC POWER CABLE RUN

Table 1

PRCS # West/Side/B Electric Power (WBEP)	Location
43	East of Bldg. 011, HH9; 30 inch cast iron lid
44	North of Bldg. 011, HH9
45	North of Bldg. 011, HH1
46	West of Building 011A
47	Not in service
48	Not in service
49	Not in service
50	Southeast corner of Bldg. 009 parking lot
51	Not in service
52	Not in service

USDA/BARC PRCS INVENTORY
WEST SIDE/AREA B
LOW VOLTAGE ELECTRIC POWER MANHOLES

Table 2

WEST SIDE/B Low Voltage Electric Power Manholes	Location
1	Manhole by substation near Bldg. 014
2	At east end of this substation near street

USDA/BARC PRCS INVENTORY
WEST SIDE/AREA B
TRANSFORMER VAULTS

Table 3

WEST SIDE/B Transformer Vaults	Location
1	In front of Bldg. 003; with open metal grate
2	In front of Bldg. 002; open metal grate

USDA/BARC PRCS INVENTORY
WEST SIDE/AREA B/NATIONAL AGRICULTURE LIBRARY
ELECTRIC POWER CABLE RUN

Table 4

PRCS # West Side/ B/Library (WBLEP)	Location
1*	North side of library on sidewalk in front of main entrance; 30 inch diameter cast iron lid
2*	between Route 1 and dirt road, near telephone pole; 30 inch diameter cast iron lid
3	between Route 1 and dirt road and south of #2 and in line with Bldg. 001; 30 inch diameter cast iron lid

* Two manhole between above two, next to each other

USDA/BARC PRCS INVENTORY
EAST SIDE/BUILDING 309 AREA
ELECTRIC POWER CABLE RUN

Table 5

PRCS # East Side/ 309/Electric Power	Location
1	East of Bldg. 309 Substation, 26 inch cast iron lid
2	West of Bldg. 309 Substation
3	In field east of Center Road
4	East of Center Road across from Bldg. 307 4-branch; 26 inch cast iron lid
5	East of Bldg. 306
6	South of Bldg. 308 parking lot
7	East of Bldg. 308; 24 inch diameter cast iron lid
8	East of Building 308C

USDA/BARC PRCS INVENTORY
EAST SIDE/ NEW SEWAGE TREATMENT PLANT
ELECTRIC POWER CABLE RUN

Table 6

East Side/STP/Electric Power (ESTPEP)	Location
1	In service driveway running south from Bldg. 218G; by 218G
2	In service driveway running south from Bldg. 218G, near Bldg. 2181
3	In service driveway running south from Bldg. 218G; at southern end of aeration lagoons

USDA/BARC STORM DRAIN PRCS INVENTORY
WEST SIDE/AREA B

Table 1

PRCS # West /B/Storm Drain (WBSD)	Location
WBSD1	NW corner of Bldg. 005 (rear of building)
WBSD2	In parking lot behind (west side of) Bldg. 005, west of SD1
WBSD3	Not in Service
WBSD4	On North Drive across from (north of) Range 1, HH6
WBSD5	On NW corner of Bldg. 004; sprayed green
WBSD5A	Not in service
WBSD5B	Trough drain just north of Bldg. 004 loading dock
WBSD5C	Trough drain off SW corner of Bldg. 004 next to sidewalk
WBSD7	SW corner of Bldg. 002
WBSD	On the southwest side of Bldg. 003
WBSD8	On SW corner of Bldg. 002 across (south of) sidewalk near tree
WBSD9	On east side of Bldg. 002 near front entrance
WBSD10	On NW corner of Bldg. 001 next to SS#9
WBSD11	On SE corner of Bldg. 006 near sidewalk and parking area
WBSD12	On NW corner of Bldg. 006 just east of entrance
WBSD13	Just west of SS#12 on west side of Bldg. 007; cover painted green

USDA/BARC STORM DRAIN PRCS INVENTORY
WEST SIDE/AREA B

Table 1

PRCS # West /B/Storm Drain (WBSD)	Location
WBSD14	Just north of SS#14 (off SW corner of parking lot between Bldg. 006 and 010A near tree) on SW end of parking lot;
WBSD14A	Trough drain just north of SS#15 near transformers for 010A
WBSD15	SE corner of North Drive and 2nd St. by Range 1, HHS
WBSD15A	At intersection of 3rd St. and Range 3 Drive by street sign; cover painted green
WBSD16	Not in service
WBSD17	Behind (SE) corner of Range 1, GH9
WBSD18	Not in service
WBSD19	Not in service
WBSD20	Not in service
WBSD21	Not in service
WBSD22	At intersection of Range 2 & 3rd Street
WBSD23	At intersection of Range 2 & 4th Street
WBSD24	In 4th Street west of (behind) Bldg. 047A
WBSD25	In 4th Street between Bldg. 014 & 048
WBSD26	At southern end of 4th Street
WBSD27	In Range 3 Orive across from Bldg. 014

USDA/BARC STORM DRAIN PRCS INVENTORY
WEST SIDE/AREA B

Table 1

PRCS # West /B/Storm Drain (WBSD)	Location
WBSD28	Not in service
WBSD29	In intersection of 2nd St. and Range 3 Drive
WBSD30	In front of (north of) Range 3, HH1
WBSD31	In front of (north of) Range 3, HH3
WBSD32	In intersection of Range 3 Drive & 3rd Street
WBSD33	In 3rd Street between Range 3, HH6 and Bldg. 009A
WBSD34	In 3rd Street just north of intersection of 3rd St. and South Drive
WBSD35	Behind (south of) Range 3, GH4
WBSD36	Not in service
WBSD37	Not in service
WBSD38	In 5th St. just off North Drive
WBSD39	Not in service
WBSD40	In field across from Bldg. 013
WBSD41	In field across from Bldg. 013
WBSD42	In field near bridge on Little Paint Road, north of North Drive
WBSD43	In field near bridge on Little Paint Road, north of North Drive
WBSD44	Just across (south of) #43 above, and behind Bldg. 013
WBSD45	In field near bridge on Little Paint Road, north of North Driveft. deep

USDA/BARC STORM DRAIN PRCS INVENTORY
WEST SIDE/AREA B

Table 1

PRCS # West /B/Storm Drain (WBSD)	Location
WBSD28	Not in service
WBSD29	In intersection of 2nd St. and Range 3 Drive
WBSD30	In front of (north of) Range 3, HH1
WBSD31	In front of (north of) Range 3, HH3
WBSD32	In intersection of Range 3 Drive & 3rd Street
WBSD33	In 3rd Street between Range 3, HH6 and Bldg. 009A
WBSD34	In 3rd Street just north of intersection of 3rd St. and South Drive
WBSD35	Behind (south of) Range 3, GH4
WBSD36	Not in service
WBSD37	Not in service
WBSD38	In 5th St. just off North Drive
WBSD39	Not in service
WBSD40	In field across from Bldg. 013
WBSD41	In field across from Bldg. 013
WBSD42	In field near bridge on Little Paint Road, north of North Drive
WBSD43	In field near bridge on Little Paint Road, north of North Drive
WBSD44	Just across (south of) #43 above, and behind Bldg. 013
WBSD45	In field near bridge on Little Paint Road, north of North Drive

USDA/BARC STORM DRAIN PRCS INVENTORY
WEST SIDE/AREA B/LIBRARY

Table 2

PRCS # West /Library/ Storm Drain (WBSD)	Location
WLSD1	Storm manhole with cast iron lid on SE side of Bldg. near sidewalk from parking lot to front entrance; on SW side of sidewalk;
WLSD2	Storm manhole with cast iron lid on SW corner of Bldg. Near loading dock and by fire hydrant
WLSD3	Trough drain on SW corner of Bldg. Right next to loading dock
WLSD4	Trough drain near oil tanks on north side of bldg.
WLSD5	On west side of bldg. is concrete pit covered with a steel grate;
WLSD6	Trough drain on SW side of library at north corner of parking lot
WLSD7	Trough drain on west side of library at intersection of roads leading from parking lot

USDA/BARC SEWER SYSTEM PRCS INVENTORY
EAST SIDE/AREA B
ANIMAL PARASITOLOGY UNIT/AREA A

Table 1

PRCS # West /B/Storm Drain (WBSD)	Location
EASS1	Not in service
EASS2	By corner of Bldg. 1043;
EASS3	By Bldg. 1104 by road
EASS4	NE corner of Bldg. 1160
EASS5	SE corner of Bldg. 1160
EASS6	in curve of road up (west) from Bldg. 1160
EASS7	In field
EASS8	In road
EASS9	In front of Bldg. 1237 door
EASS11	Not in service
EASS12	Not in service
EASS13	Not in service
EASS14	Storm drain in road across from Bldg. 1302 and near field
EASS15	Near Bldg. 1320 and close to field

USDA/BARC SEWER SYSTEM PRCS INVENTORY
EAST SIDE/AREA B
ANIMAL PARASITOLOGY UNIT/AREA A

Table 1

PRCS # West /B/Storm Drain (WBSD)	Location
EASS16	On southern side of Bldg. 1280 near gated entrance
EASS17	Near Bldg. 1280 and directly across the street from Bldg. 1320
EASS18	In street by pumping station 1286
EASS19	Storm drain between pumping station 1286 and Bldg. 1287
EASS20	By side of road between Bldgs. 1287 & 1288
EASS21	Not in service
EASS22	By Bldg. 1290
EASS23	Not in service
EASS24	By Bldg. 1017; cover labeled steam
EASS25	In gravel road by Bldg. 1017
EASS26	Manhole in road next to pumping station 100G
EASS27	Not in service
EASS28	Not in service
EASS29	Not in service
EASS30	Not in service

USDA/BARC SEWER SYSTEM PRCS INVENTORY
EAST SIDE/AREA B
ANIMAL PARASITOLOGY UNIT/AREA A

Table 1

PRCS # West /B/Storm Drain (WBSD)	Location
EASS31	Storm drain; continuing east in field near oak tree and Bldg. 1043
EASS32	Not in service
EASS33	Not in service
EASS34	Not in service
EASS35	Not in service
EASS36	Not in service
EASS37	About 75 ft. Behind Bldg. 1062
EASS38	About 75 ft. Behind Bldg. 1063
EASS39	About 75 ft. Behind Bldg. 1064
EASS40	About 75 ft. behind and between Bldgs. 1064 and 1065
EASS41	Not in service
EASS42	About 75 ft. Behind Bldg. 1065
EASS43	Not in service
EASS44	Behind Bldg. 1120 next to evergreen bush
EASS45	Not in service

USDA/BARC SEWER SYSTEM PRCS INVENTORY
EAST SIDE/AREA B
ANIMAL PARASITOLOGY UNIT/AREA A

Table 1

PRCS # West /B/Storm Drain (WBSD)	Location
EASS46	Not in service
EASS47	Behind Bldg. 1082 and near Bldg. 1081
EASS48	Behind Bldg. 1082
EASS49	Near water tower by these Bldgs. 1080-1083
EASS50	Near left side of front entrance of Bldg. 1145
EASS51	On right side of Bldg. 1145; cover marked sewer
EASS52	On right side of Bldg. 1146; 4 ft deep
EASS53	On right side of Bldg. 1144;
EASS54	On right side of Bldg. 1143
EASS55	In front of Bldg. 1142
EASS56	In front of Bldg. 1141
EASS57	On left side of driveway leading to Bldg. 1140
EASS58	Behind Bldg. 1140 and in line with EASS57 and in line with EASS40 (near water tower)
EASS59	Behind Bldg. 1104
EASS60	In road right in front of Bldg. 1190 entrance; marked sewer; no inlet from Bldg. 1190

USDA/BARC SEWER SYSTEM PRCS INVENTORY
EAST SIDE/AREA B
ANIMAL PARASITOLOGY UNIT/AREA A

Table 1

PRCS # West /B/Storm Drain (WBSD)	Location
EASS61	In parking area of Bldg. 1191 marked sewer
EASS62	In road right in front of Bldg. 1194
EASS63	Bldgs. 1060 series; behind rear of barn
EASS64	In rear of Bldg. 1080 in grass near picnic table
EASS65	Not in service
EASS66	In front of Bldg. 1381
EASS67	In front of Bldg. 1384
EASS68	Not in service
EASS69	Behind Bldg. 1423 near where pigs feed
EASS70	Behind Bldg. 1423 near red fuel tank
EASS71	To right of entrance to Bldg. 1420
EASS72	Near pumping station; bolted down cover;
EASS73	Off roadside across from pumping station and near where pigs feed
EASS74	Across from back of pig pen feeding area and near pumping station
EASS75	On SW side of pumping station in field; bolted down;
EASS76	In road down from Bldg. 1390 and on way to pumping station; paved over; between pole with transformer and pole going south towards Bldg 1390

USDA/BARC SEWER SYSTEM PRCS INVENTORY
WEST SIDE/AREA B

Table 2

PRCS # West /B/Storm Drain (WBSD)	Location
WBSS1	NW corner behind Bldg. 005 (in rear)
WBSS2	In 1st St. road behind (west of) Bldg. 005, west of SS1
WBSS3	In field north of North Dr. And about 50 yards SE of wet well (011B)
WBSS4	In North Drive road (north side of road) across from (north of) Range 1 HH6
WBSS5	Behind Bldg. 003 on NW side of bldg., east of 1st street
WBSS6	Behind Bldg. 003 on SW corner of bldg in sidewalk, east of 1st street
WBSS6A	Behind (west side of) Bldg.003 near loading dock in 1st Street
WBSS7	On south side of Range 2 Drive between parking lot and Bldg. 006 in grass (front yard for Bldg. 006)
WBSS8	In south side of Range 2 Drive and SW; cover labeled SUMP PUMP; next to sidewalk;
WBSS8A	In parking lot on west side of Bldg. 006
WBSS9	Behind Bldg. 001 NW corner by sidewalk
WBSS10	In grass on SE corner of Bldg. 002 by picnic table
WBSS11	In parking lot south of Bldg. 006 but near SE bldg. corner;
WBSS12	On west side of Bldg. 007
WBSS13	50 feet north of #12; in parking lot for Bldg. 007; frame damaged and bent up from vehicle traffic

USDA/BARC SEWER SYSTEM PRCS INVENTORY
WEST SIDE/AREA B

Table 2

PRCS # West /B/Storm Drain (WBSD)	Location
WBSS14	West side of intersection of Bldg. 006 parking lot and Terrace Drive;
WBSS15	SW corner of parking lot between Bldg. 006 and 010A near tree
WBSS16	East of but near intersection of Range 3 Drive and 2nd Street and NE of steam manhole
WBSS17	Intersection of Range 3 Drive and 2nd Street
WBSS18	Between Range 3 HH3 & HH4
WBSS19	South of Range 3 Drive & 3rd Street intersection
WBSS20	In front or east of Bldg. 009AA
WBSS21	North of South Drive and 3rd St. intersection; cover says steam
WBSS22	Just west of #21 in grass
WBSS23	Northeastern corner of Bldg. 050 just south of bldg. sign
WBSS24A	Sewer between 2nd and 3rd set of entrances on north side of Bldg. 050
WBSS25	In front of (on north side of) Bldg. 050 western end, near cupola
WBSS26	South of intersection of 3rd St. & South Drive, next to fire hydrant
WBSS27	Directly south of intersection of 3rd. St. & South Drive
WBSS28	North of 2nd Street and east of Bldg. 010A

USDA/BARC SEWER SYSTEM PRCS INVENTORY
WEST SIDE/AREA B

Table 2

PRCS # West /B/Storm Drain (WBSD)	Location
WBSS28	North of 2nd Street and east of Bldg. 010A
WBSS29	In Range 2 Drive behind (south of) Range 1 GH11
WBSS30	Southeastern corner of Range 1 GH17 in grass (storm drain)
WBSS31	In Range 2 Drive behind (south of) Range 1 GH17
WBSS32	In Range 2 Drive behind (south of) Range 1 GH18
WBSS33	In Range 2 Drive on northwest side of Bldg. 046A
WBSS34	Behind (west of) Bldg. 047A
WBSS35	Behind (west of) Bldg. 049A and near Bldg. 048
WBSS36	West of 4th Street and in line with WBSS34 and west of Bldg. 014; about 20 feet northwest of aboveground oil tanks
WBSS37	Not in service
WBSS38	In parking lot on southwest side of Bldg. 011A, near steps to building entrance
WBSS39	In parking lot southwest of Bldg. 011A and across from Range 1, HH13
WBSS40	Not in service
WBSS41	Not in service
WBSS42	On west side of East Line Road (ELR) near ELR intersection with and North Farm Road
WBSS43	Just south of #42

USDA/BARC SEWER SYSTEM PRCS INVENTORY
EAST SIDE/SECURITY AREA

Table 3

Manhole # ESECSS	Location
ESECSS1	Not in service
ESECSS2	Not in service
ESECSS3	Not in service
ESECSS4	Continuing east but across (east side) of Dairy Road near Pumping Station 194
ESECSS5	Not in service

USDA/BARC SEWER SYSTEM PRCS INVENTORY
EAST SIDE/BUILDING 192E AREA

Table 4

Manhole # E192SS	Location
E192SS1	Next to (north side of) Pumping Station (Bldg. 194) in field
E192SS2	North of #1 in field north of Pumping Station
E192SS3	Not in service
E192SS4	Not in service
E192SS5	Not in service
E192SS6	Not in service
E192SS7	Not in service
E192SS8	Not in service
E192SS9	Not in service
E192SS10	Not in service
E192SS11	Not in service

USDA/BARC SEWER SYSTEM PRCS INVENTORY
EAST SIDE/DAIRY/AREA D

Table 5

Manhole EDSS#	Location
EDSS1	Not in service
EDSS2	Not in service
EDSS3	Not in service
EDSS4	Right next to EDSS5; by road that runs along Bldgs. 171 and 166
EDSS5	West of E192SS2 on road running east/west along Bldgs. 171 & 166
EDSS6	West of EDSS5 and on road running east/west along Bldgs. 171 and 166
EDSS7	West of EDSS6; off SW corner of Bldg. 171
EDSS8	In road running along (south of) Bldgs. 166 and 171
EDSS9	Just east of intersection of South Dairy Road and road above
EDSS10	Just west of South Dairy Road and off SW corner of Bldg. 168
EDSS11	Due west of EDSS10 and off SE corner of Bldg. 168
EDSS12	Northwestern side of Bldg. 167 in road and across from Bldg. 161
EDSS13	NE Corner of Bldg. 163A; brick construction; 26 inch cast iron lid
EDSS14	NW corner of Bldg. 163A
EDSS15	On south side and between Bldgs. 162 &164

USDA/BARC SEWER SYSTEM PRCS INVENTORY
EAST SIDE/DAIRY/AREA D

Table 5

Manhole EDSS#	Location
EDSS16	In road on south side of (in front of) Bldg. 165
EDSS17	Storm drain west of Bldg. 177 in road and across from Bldg. 177
EDSS18	Across from Bldg. 177 in road just a little south of storm drain above
EDSS19	Storm drain on south side of Bldg. 176
EDSS20	On south side of Bldg. 176 in road; just a little south of storm drain
EDSS21	SW side of Bldg. 173; rectangular manhole
EASS22	About 30 feet south of Bldg. 177 in roadway
EASS23	About 10 feet west of roadway off SW corner of 1778
EASS24	Sump pit 15 yards north of Bldg. 161
EASS25	SW corner of Bldg. 163F in intersection of South Farm Road
EASS26	On the north side of Bldg. 163F between Bldg. 163F & 1638
EASS27	On the north side of Bldg. 163 a little west of center
EASS28	On the southwest corner of Heifer lot 1
EASS29	Not in service
EASS30	About 25 yards from the NE corner of Bldg. 178-2

USDA/BARC SEWER SYSTEM PRCS INVENTORY
EAST SIDE/BUILDING 215 AND NEARBY AREA

Table 6

Manhole # E215SS	Location
E215SS1	South of ECSS17; southeast of Dairy and north of Beaver Dam Road
E215SS2	Runs south and crosses Beaver Darn Road
E215SS3	West of Bldg. 215 in field is cast iron lid on concrete base
E215SS4	South side of Bldg. 215 and between 2158 and 21SC
E215SS5	South of E215SS4
E215SS6	South of E215SS5 and north of old sewage treatment plant
E215SS7	Older line next to and connected to E215SS6
E215SS8	Manhole by pumping station 218

USDA/BARC SEWER SYSTEM PRCS INVENTORY
EAST SIDE/POULTRY/AREA C

Table 7

Manhole# ECSS	Location
ECSS1	Not in service
ECSS2	Not in service
ECSS3	Off NW corner of Bldg. 254; Cover labeled Sewer
ECSS4	Not in service
ECSS5	Not in service
ECSS6	Not in service
ECSS7	Not in service
ECSS8	On southern side (in front of) Bldg. 267 in field across from driveway
ECSS8A	Not in service
ECSS9	On west side (behind) Bldg. 262 near back entrance; manhole with cast iron lid
ECSS10	Off NW corner of Bldg. 264 is manhole with cast iron lid
ECSS11	Off northern side of Bldg. 264 near air-handling unit is manhole with cast iron lid
ECSS12	Not in service
ECSS13	Not in service
ECSS14	Not in service

USDA/BARC SEWER SYSTEM PRCS INVENTORY
EAST SIDE/POULTRY/AREA C

Table 7

Manhole# ECSS	Location
ECSS15	Not in service
ECSS16	Not in service
ECSS17	In wooded area NE of ECSS16 and NE of Poultry Road, just as it begins to curve east, heading north to Poultry area
ECSS18	In wooded area north of ECSS17 and south of Bldg. 236
ECSS19	North of ECSS18 in wooded area behind (south of) Bldg. 236
ECSS20	Behind (just off SE corner of Bldg. 236) near fence; brick and concrete upraised base; manhole lid labeled Sewer
ECSS21	On SE side of (in front of) Bldg. 262 in sidewalk is manhole with cast iron lid
ECSS22	Behind (west of) Bldg. 265
ECSS23	Behind (west of) Bldg. 264
ECSS24	Northwest corner of Bldg. 263
ECSS25	Near SS13 but closer to parking lot; upraised concrete base with cast iron lid
ECSS26	Not in service
ECSS27	Off SW corner of Bldg. 254; upraised concrete base; cover labeled Sewer
ECSS28	Across the (on west side of) road across from Bldg. 255A is upraised concrete manhole with cast iron lid
ECSS29	On NE side of Bldg. 236 but across from entrance road in grass; 6 feet deep; rusty access rungs

USDA/BARC SEWER SYSTEM PRCS INVENTORY
EAST SIDE/POULTRY/AREA C

Table 7

Manhole# ECSS	Location
ECSS30	On east side of Bldg. 236 but just south of the road intersection; 28 inch cast iron lid
ECSS31	Behind (South side of)Bldg. 236 near back entrance; manhole with cast iron lid;
ECSS32	On SW corner of Bldg. 236 manhole with cast iron lid
ECSS33	In woods east of Poultry Road
ECSS34	Not in service
ECSS35	Not in service
ECSS36	West of Poultry Road and north of intersection of Poultry Road and road to Bldg. 226
ECSS37	West of Poultry Road and south of intersection of Poultry Road and road to Bldg. 226
ECSS38	Not in service
ECSS39	Not in service

USDA/BARC SEWER SYSTEM PRCS INVENTORY
EAST SIDE/ANIMAL HUSBANDRY

Table 8

Manhole# EAHSS	Location
EAHSS1	Junction where Dairy line ties in; this line then runs south to Sewage Treatment Plant at Bldg. 218
EAHSS2	South of Powder Mill Road, west of Sheep Road
EAHSS3	South of Powder Mill Road, west of Sheep Road
EAHSS4	Just north of fence in field, and west of Sheep Road
EAHSS5	Just north of fence in field and east of Sheep Road
EAHSS6	Just north of fence in field, south of Powder Mill and east of Sheep Road
EAHSS7	Line from Poultry area ties in here; this manhole slightly Southwest of the intersection of Powder Mill/Poultry Road intersection and a few feet west of EAHSS6
EAHSS8	Line from Beef Cattle Barn (Bldg. 223) ties in here; south of the Powder Mill/Poultry Road intersection
EAHSS9	Not in service
EAHSS10	Northwest of Building 209 and runs SW to pumping station 194; farthest from pumping station
EAHSS11	Southwest of EAHSS1 and runs SW to pumping station 194
EAHSS12	Southwest of EAHSS2,runs SW to pumping station, and nearest to pumping station 194
EAHSS13	Northwest of Bldg. 203 sewer
EAHSS14	Not in service
EAHSS15*	S S manhole in parking lot east of 201A and south of Bldg. 202

* SS (Sewer System)

USDA/BARC SEWER SYSTEM PRCS INVENTORY
EAST SIDE/ANIMAL HUSBANDRY

Table 8

Manhole# EAHSS	Location
EAHSS16	S S manhole in parking lot NE of Bldg. 202
EAHSS17	South of Bldg. 203
EAHSS18	It is across the street from SW corner of Bldg. 204, and near above ground tank storage
EAHSS19	South of Bldg. 204; 26 inch cast iron lid; upraised base
EAHSS20	Southeast of Bldg. 204 in cow pasture; again upraised base
EAHSS21	East of EAHSS20 and close to poultry road; in cow pasture
EAHSS22	Northeast corner of Bldg. 2048; brick construction
EAHSS23	Due north of EAHSS7 in field
EAHSS24	Northwest corner of Bldg. 204 B; 6 inch cast iron pipe; brick construction
EAHSS25	Southeast corner of Bldg. 204B
EAHSS26	Off southwest corner of Bldg. 2048, but across the little road
EAHSS27	Southeast of Bldg. 211 is manhole in cow pasture between 2 telephone poles; on concrete base
EAHSS28	About 2-3 feet north of these 2 telephone poles is another manhole
EAHSS29	About 10 feet southwest of these telephone poles is a square manhole
EAHSS30	Storm drain about 20 feet southeast side of Bldg. 200 in grass; brick and concrete construction with cast iron lid;

* SS (Sewer System)

USDA/BARC SEWER SYSTEM PRCS INVENTORY
EAST SIDE/ANIMAL HUSBANDRY

Table 8

Manhole# EAHSS	Location
EAHSS31	East of EAHSS30 (on southeast side of Bldg. 200); brick and concrete construction; 24 inch lid
EAHSS32	Not in service
EAHSS33	SW corner of Bldg. 204 left of entrance
EAHSS34	Not in service
EAHSS35	Drainage manhole at SE corner of Bldg 203
EAHSS36	Not in service
EAHSS37	Not in service
EAHSS38	Southeast corner of Bldg. 2048 sewer
EAHSS39	Not in service
EAHSS40	NW corner of Bldg. 203
EAHSS41	West side 208C; did not see; in pig pen; 6 inch transite sewer
EAHSS42	Southwest corner of 203A
EAHSS43	Slightly SW of previous manhole at NW corner of Bldg. 203
EAHSS44	Manhole on east side of 203C; drainage for sheep barn
EAHSS45	West of EAHSS44 and southwest of Bldg. 203C
EAHSS46	West of EAHSS45 in field

* SS (Sewer System)

USDA/BARC SEWER SYSTEM PRCS INVENTORY
EAST SIDE/BUILDING 223/BEEF BARN

Table 9

Manhole# EBBSS	Location
EBSS1	In cow pasture SW of Bldg. 223 is manhole with upraised base and cast iron lid

USDA/BARC SEWER SYSTEM PRCS INVENTORY
EAST SIDE/FDA

Table 10

Manhole# EFDASS	Location
EFSASS1	Not
TO	In
EFSASS40	Service

USDA/BARC SEWER SYSTEM PRCS INVENTORY
EAST SIDE/NUTRITION

Table 11

Manhole# ENUTSS	Location
ENUTSS1	NE corner of Bldg. 309 in parking lot; cover labeled Sewer ST
ENUTSS2	NE corner of Bldg. 309; covered labeled storm sewer
ENUTSS3	Storm drain; SE side of Bldg. 308 in grass just east of sign (Beltsville Human Nutrition Research Center); brick and concrete rectangular pit with 2.5 ft x 2.5 ft diamond plate steel lid; in line with outfall stream running towards boiler plant;
ENUTSS4	SW corner of Bldg. 308C; 2.5 ft x 2.5 ft diamond plate steel lid over concrete pit; storm drain
ENUTSS5	Not in service

USDA/BARC SEWER SYSTEM PRCS INVENTORY
EAST SIDE
RSS/VISITOR CENTER/INSTRUMENTATION

Table 12

RSS

Manhole# EFOSS	Location
EFOSS1	In grass on SW corner of Bldg. 301; cover labeled S;
EFOSS2	Just south of Bldg. 301 equipment yard near back entrance; cover labeled Sewer;
EFOSS3	On east side of Bldg. 301D

Building 302-Visitor Center and Building 303-Instrumentation

Manhole# EVISS	Location
EVISS1	On west side (behind) Visitor Center near AHU
EVISS2	On southeast corner (in front of) Visitor Center
EVISS3	Just east of Visitor Center Road and about 15 ft. South of Powder Mill; in grass; manhole cover says S
EVISS4	Off NE corner of Bldg. 303 in grass

USDA/BARC SEWER SYSTEM PRCS INVENTORY
EAST SIDE/WATER TREATMENT

Table 13

Manhole# EWTSS	Location
EWTSS1	On NE corner of Bldg. 310; 10-12 ft deep; 26 in diameter cast iron lid with the letter D on it; upraised base
EWTSS2	Heading 15-20 south of manhole above (#1) and behind (east side of) Bldg 310; this manhole is not upraised; cast iron lid with the letter D on it
EWTSS3	50 feet east of the SE corner of Bldg 310

USDA/BARC SEWER SYSTEM PRCS INVENTORY
EAST SIDE
ENTOMOLOGY AREA/PESTISIDE AREA

Table 14

Manhole# EENTSS	Location
ENTSS1	South west of Powder Mill Road and Pesticide Road intersection
ENTSS2	24 inch diameter cast iron lid on concrete base at intersection of Entomology Road (to Bee area) and Powder Mill
ENTSS3	24 inch diameter cast iron lid on concrete base north of ENTSS2 and just south of intersection of Entomology Road and Radio road leading to Bldgs. 310,309 etc.
ENTSS4	24 inch diameter cast iron lid on concrete base north of ENTSS3; just north of Entomology and Radio Roads intersection
ENTSS5	24 inch diameter cast iron lid on concrete base north of ENTSS4; sewer line from 426 shops joins the line here to go to pumping station behind fitness center
ENTSS6	24 inch diameter cast iron lid on concrete base north of ENTSS5
ENTSS7	Not in service
ENTSS8	Not in service
ENTSS9	Not in service
ENTSS10	NE corner of Bldg. 483; in grass; 24 inch diameter cast iron lid
ENTSS11	South (behind) of Bldg. 476 in grass; 24 inch diameter cast iron lid
ENTSS12	SW corner of Bldg. 476 in road; 24 inch diameter cast iron lid
ENTSS13	NW corner of Bldg. 476 in road; 24 inch diameter cast iron lid
ENTSS14	Not in service
ENTSS15	Along west side of service road and across from Bldg. 465

USDA/BARC SEWER SYSTEM PRCS INVENTORY
EAST SIDE
ENTOMOLOGY

Table 15

Manhole# EENTSS	Location
ENTSS16	SE corner of Bldg. 465 in field; 24 inch diameter lid
ENTSS17	NE corner of greenhouse in field; 24 inch diameter lid; lid spray-painted fluorescent green
ENTSS18	Not in service
ENTSS19	NW corner of Bldg. 467; 24 inch diameter lid with S on cover
ENTSS20	The following series of manholes are located on the east side of the service road going to the Entomology area; this line runs south along the west side of this road and ties in with the #6 line from Zoology; at this point the line (#4 line) then continues south to Bldg. 304 pumping station
ENTSS21	South of above manhole
ENTSS22	South of above manhole
ENTSS23	South of above manhole
ENTSS24	South of above manhole
ENTSS25	South of above manhole
ENTSS26	South of above manhole

USDA/BARC SEWER SYSTEM PRCS INVENTORY
EAST SIDE/PESTISIDE AREA

Table 16

Manhole# EENTSS	Location
EPESTSS1	Not in service
EPESTSS2	Not in service
EPESTSS2A	Not in service
EPESTSS7	Not in service
EPESTSS8	Not in service
EPESTSS9	Not in service
EPESTSS9A	Not in service
EPESTSS10	Not in service

USDA/BARC SEWER SYSTEM PRCS INVENTORY
EAST SIDE/FITNESS

Table 1

Note: Tanks are underground unless otherwise noted

Manhole# EFITSS	Location
EFITSS1	24 inch diameter cast iron lid on east side of Bldg. 426 near entrance from Powder
EFITSS2	SW of Bldg. 421 entrance; 24 inch cast iron lid on upraised concrete base; lid has lift handle
EFITSS3	Near service road and across the way from south side of Bldg. 421; 24 inch diameter cast iron lid with rusty lift handle
EFITSS4	Not in service
EFITSS5	Not in service
EFITSS6	Not in service
EFITSS7	Not in service
EFITSS8	Not in service
EFITSS9	This series of manholes from EFITSS9 to EFITSS13 is in the woods behind Bldgs. 421 to 304
EFITSS10	In wooded area behind 425 heading toward the pumping station Bldg. 304
EFITSS11	In wooded area behind 425 heading toward the pumping station Bldg. 304
EFITSS12	In wooded area behind 425 heading toward the pumping station Bldg. 304.
EFITSS13	In wooded area behind 425 heading toward the pumping station Bldg. 304

USDA/BARC PRCS INVENTORY
Fuel Tanks

Table 1
Note: Tanks are underground unless otherwise noted

Location	Tank
Bldg. 027	4000 gallon gasoline
Bldg. 014	75,000 gallon #2 fuel oil
Bldg. 014	75,000 gallon #2 fuel oil
Bldg. 165	Not in service
Bldg. 166	4000 gallon #2 fuel oil
Bldg. 166	2500 gallon #2 fuel oil
Bldg. 202	20,000 gallon #2 fuel oil
Bldg. 261	6000 gallon #2 fuel oil
Bldg. 445	10,000 gallon #2 fuel oil
Bldg. 309	20,000 gallon #2 fuel oil
Bldg. 309	20,000 gallon #2 fuel oil
Bldg. 010	Not in service
Bldg. 186 (Security)	Not in service
Bldg. 215	Not in service
Bldg. 470	Not in service

USDA/BARC PRCS INVENTORY
Fuel Tanks

Table 1

Note: Tanks are underground unless otherwise noted

Location	Tank
Bldg. 407	Not in service
Bldg. 208	Not in service
Bldg. 236	Not in service
Bldg. 320A	Not in service
Library	2 underground oil tanks on NW side
Bldg. 327A	Not in service
Bldg. 303	Changed to Natural Gas
Bldg. 426	Changed to Natural Gas
Bldg. 431	Changed to Natural Gas
FDA area	Not in service
FDA incinerator	Propane tank (above ground)
Bldg. 1160	#2 fuel oil
Bldg.1001	#2 fuel oil

U. S. DEPARTMENT OF AGRICULTURE BELTSVILLE AGRICULTURAL RESEARCH CENTER

Miscellaneous Spaces Inventory of Confined Spaces

This report is a summary of miscellaneous confined spaces which involved only the locating of such spaces. Some of the spaces appear to be valve shut-off vaults. However, the following spaces were not evaluated nor characterized under a specific system such as sanitary, storm, etc. These spaces are assumed to be permit-required confined spaces until they are further evaluated. These confined spaces are summarized in Table 1.

1.0 Hazards

The principal hazards in these spaces are assumed to consist of the following:

a. Engulfment hazard (some spaces were filled with water)

b. Oxygen deficiency and buildup of hazardous atmosphere, including buildup of explosive methane gas

c. Slip, trip and fall hazard

d. Rectangular vaults that have 2 steel plate covers can be a hazard when only one plate is removed for entry. The other plate can be easily dislodged and fall into the space while a worker is in there.

e. All of these spaces meet the OSHA definition of a permit-required confined space (PRCS) until further evaluation had occurred

2.0 Recommendations

a. Work in these spaces confined space entry and appropriate steps must be taken to comply with the OSHA Permit-Required Confined Space Standard (1910.146), including, but not limited to:

1) Entry permit

2) Air monitoring for LEL, oxygen, carbon monoxide, and hydrogen sulfide

3) Ventilation of the space

4) Emergency procedures and rescue

5) Fall protection and retrieval for the water towers and those vaults or pits greater than 5 feet in depth

6) Isolation of all hazardous energy sources including electrical and hydraulic (fluids under pressure in piping)

7) A minimum of 2 people for each entry (entrant and attendant/entry supervisor)

b. Work in these confined spaces may require the use of personal protective equipment such as eye and face protection, protective gloves, and head protection. This personal protective equipment (PPE) must meet the following OSHA requirements:

1910.132 - General PPE requirements including selection, use and training

1910.133- Eye and Face Protection

1910.135 - Head Protection

1910.136 - Foot Protection

c. Manholes may be inhabited by biological hazards. Workers should be aware of biological hazards such as snakes, insects, and other animals.

USDA/BARC PRCS INVENTORY
MISCELLANEOUS CONFINED SPACES

Table 1

PRCS#	Location
1	Near parking lot north of Bldg. 308; 2.5 ft x 2.5 ft diamond plate steel lid on rectangular concrete
2	Not in service
3	Not in service
4	Not in service
5	Not in service
6	Not in service
7	Not in service
8	Not in service
9	Not in service
10	Not in service
11	Not in service
12	Not in service
13	Not in service
14	Directly south of Bldg. 211 near entrance road to this area and across from southeast corner of Bldg. 200
15	On the southeast side of Bldg. 200 in grass, just west side of entrance road; 26 inch cast iron lid; lid says sewer
16	On east side of Bldg. 200 by sidewalk with 26 inch cast iron lid; cover says sewer

U. S. DEPARTMENT OF AGRICULTURE BELTSVILLE AGRICULTURAL RESEARCH CENTER

Ventilation and HVAC Systems/Elevator Pits Inventory of Confined Spaces

1.1 Background

a. HVAC System

The HVAC systems of various buildings were not inspected at this time. Fan units that have access doors with such dimensions that allow employees enter and exit the mixing chamber, heating and cooling coils, and filter banks on foot with no difficulty are not considered permit-required confined spaces. There are no atmospheric hazards in these spaces and they do not require warning signs. Entry into these spaces is covered by the lockout/tagout program.

Duct access hatches are 1 square foot in size. The ducts cannot be accessed through these hatches and are therefore not confined spaces. Should ducts be cut open for access so that workers can enter the duct, then entry must be done as per the permit- required confined space program.

b. Carpentry Shop, Building 426

The wood dust collector and exhaust fan serving the carpentry shop is located above and just east of the entrance to the carpentry shop. Entry into the dust collector is possible by removal of a 2 foot-square access panel. This space is a permit-required confined space. The following is a summary of hazards when entering this space:

1) Wood dust may create an explosive dust atmosphere

2) The space slopes inward and a worker may become trapped

3) There are mechanical and electrical hazards from equipment in the dust collector and fan unit.

c. Elevator Pits

Elevator pits meet the OSHA definition of a confined space. There are no atmospheric hazards in these spaces and they do not require warning signs. Entry into these spaces in order to do servicing and maintenance work is covered by the lockout/tagout program.

d. Walk-in Refrigerators

Walk-in refrigerators were not inspected, but were described by BARC personnel. The refrigerators have access doors with such dimensions that allow employees to enter and exit on foot with no difficulty. These walk-in refrigerators are not considered permit-required confined spaces. Entry and work into these spaces is covered by other applicable OSHA standards, including but not limited to, the Emergency Action Plan,(1910.38), Emergency Egress (1910.36 and 1910.37), and the lockout/tagout\ program. Exposures to potentially hazardous atmospheres are covered by OSHA's Toxic and Hazardous Substances standards (Subpart Z - 1910.1000).

USDA/BARC PRCS INVENTORY
EAST SIDE NEW SEWAGE TREATMENT PLANT
(BUILDING 218)

Table 1

PRCS#	Location
Comminutor	Building 218, Influent pumping station
Wet well	Building 218, Influent pumping station
Dry well	Building 218, Influent pumping station
Clarifier splitter box	Building 218
Clarifier	Building 218
Clarifier	Building 218
Return sludge pumps wet and dry well	Building 218
Scum pit	Building 218
Mixed Liquor Station wet and dry well	Building 218
Effluent filter tanks	Building 218
Contact chamber	Building 218
Effluent Splitter box	Building 218
Effluent pumping station	Building 218
Final contact chamber	Building 218E
Sludge holding tank	Building 218
Sludge holding tank	Building 218

USDA/BARC PRCS INVENTORY
EAST SIDE OLD SEWAGE TREATMENT PLANT

Table 2

PRCS#	Location
Wet well	West of new sewage treatment plant
Primary clarifier (round)	Not in service
Primary clarifier (square)	Not in service
Trickling filter	Not in service
Secondary clarifier (round)	Not in service
Secondary clarifier (round)	Not in service
Secondary clarifier (square)	Not in service
Chlorine contact chamber	Not in service
Sludge holding tank	Not in service
Anaerobic digester	Not in service
Anaerobic digester	Not in service

USDA/BARC PRCS INVENTORY
WEST SIDE SEWAGE TREATMENT PLANT
(BUILDING 015)

Table 3

PRCS#	Location
Influent wet well	Building 015
Influent dry well	Building 015
Influent comminutor	Not in service
Flow equalization tank*	Building 015
Clarifier/aeration/sludge tank*	Building 015
Clarifier/aeration/sludge tank*	Building 015
Sludge holding tank	Building 015
Sand filters chamber	Building 015
Cascading aeration/UV chamber	Building 015
Effluent pumping station wet well	Building 015
Effluent pumping station dry well	Building 015
Above-ground methanol tank	Building 015; 2000 gallon storage tank

USDA/BARC PRCS INVENTORY
WASTEWATER PUMPING STATIONS

Table 1

Pumping Station Bldg #	Location
304	Wet well 12 feet deep, 10 foot diameter
304	Dry well 12 feet deep, 10 foot diameter
308C	Wet well only 10 feet deep, 5 foot diameter; only services Bldg. 308C; no winch
194	In field north of Powder Mill Road across from Dairy; wet well
194	In field north of Powder Mill Road across from Dairy; dry well
1008	East side of APU area; pumps directly to #1500 station at APU; no winch
1500	Central section of APU area; wet well; pumps directly to Building 218 lift station
1500	Central section of APU area; dry well
1286	Western side of APU area; wet well; pumps directly to #1500 station
1286	Western side of APU area; dry well
344	Behind (east of) barn 344 at the FDA; wet well only (no winch)
0118	Wet well only; south of Bldg. 011A and only services Bldg. 011A; no winch
218	Influent pumping station for east side sewage treatment plant; wet well
218	Influent pumping station for east side sewage treatment plant; dry well
218G	Wet well only; only services Bldg. 218G; no winch

USDA/BARC PRCS INVENTORY
WEST SIDE/AREA B

Table 1

PRCS # West Side/8/Steam (WBST)	Location
WBST1A	East of HH1 off 1st St. near North Drive
WBST1	In front of entrance to HH1 and closer to North Street than its companion manhole (#2 below); 30 inch diameter cover
WBST2	In front of entrance to HH1 and closer to entrance pipes directly below opening; 22 inch diameter cover
WBST3	In front of entrance to HH2 and closer to North Street than its companion manhole (#4 below); 30 inch diameter cover
WBST4	In front of entrance to HH2 and closer to entrance; 22 inch diameter cover
WBSTS	In front of entrance to HH3 and closer to North Street than its companion manhole (#6 below); 30 inch diameter cover
WBST6	In front of entrance to HH3 and closer to entrance; 22 inch diameter cover
WBST7	In front of entrance to HH4 and closer to North Street than its companion manhole (#8 below); 30 inch diameter cover
WBST8	In front of entrance to HH4 and closer to entrance; 22 inch diameter cover
WBST9	In front of entrance to HHS and closer to North Street than its companion manhole (#10 below); 30 inch diameter cover
WBST10	In front of entrance to HH4 and closer to entrance; 22 inch diameter cover
WBST11	In front of entrance to HH5 and closer to North Street than its companion manhole (#12 below); 30 inch diameter cover
WBST12	In front of entrance to HH5 and closer to entrance; 22 inch diameter cover
WBST13	In front of entrance to HH6 and closer to North Street than its companion manhole (#14 below); 30 inch diameter cover
WBST14	In front of entrance to HH6 and closer to entrance; 22 inch diameter cover

USDA/BARC PRCS INVENTORY
WEST SIDE/AREA B

Table 1

PRCS # West Side/8/Steam (WBST)	Location
WBST15	In front of entrance to HH7 and closer to North Street than its companion manhole (#16 below); 30 inch diameter cover
WBST16	In front of entrance to HH7 and closer to entrance; 22 inch diameter cover
WBST17	In front of entrance to HH8 and closer to North Street than its companion manhole (#18 below); 30 inch diameter cover
WBST18	In front of entrance to HHS and closer to entrance; 22 inch diameter cover
WBST19	At western end of HHS; valve pit closer to North Street than its companion manhole (#20 below); 30 inch diameter cover; manhole covered marked S
WBST20	At western end of HH8 and closer to building; 30 inch diameter cover
WBST21	Between HH9 & HH10 in grass under window, 24 inch diameter cover
WBST22	Near #21 (above) and between HH9 &HH10 in sidewalk; contains main steam line; expansion joint; 24 inch diameter cover
WBST23	Located to the right of HH10 entrance in sidewalk
WBST24	Located on the northern side of #23 (above)
WBST25	Located in sidewalk across from HH12 entrance; 30 inch diameter cover
WBST26	Located between HH13 &HH14; circular manhole cover with open grate
WBST27	In front of HH14 entrance on sidewalk;
WBST28	In sidewalk in front of HH14 between HH 15 & HH16, closer to North Drive.; 24 inch diameter cover
WBST29	In sidewalk between HH16 & HH17; 24 inch diameter cover

USDA/BARC PRCS INVENTORY
WEST SIDE/AREA B

Table 1

PRCS # West Side/8/Steam (WBST)	Location
WBST30	In front of HH17
WBST31	In grass between HH17 & HH18 flush with wall
WBST32	Cupola NE corner of Range 3 Dr. & 3rd St.
WBST33	Manhole next to cupola; 3 x 3 rectangular metal cover on concrete base; NE corner of Range 3 Dr. & 3rd St.; only entrance to large steam tunnel run from Bldg 014
WBST34	Manhole at 4th St. and Range 3 Drive intersection in street
WBST35	South of #33 above; at 4th St. and Range 3 Drive
WBST36	Not in service
WBST37	Cupola at Range 1 Drive and rear (south side of) GH18
WBST38	Manhole right behind cupola 35A by ramp entrance to greenhouse
WBST39	On west side of Range 2, HH5 near 3rd St. stop sign; brick construction
WBST40	Between HHS and GH5 near 3rd St. and across the street from 047A; entrance via ladder to Range 2 tunnel
WBST41	In front of (east side) of Bldg. 046
WBST42	West entrance Range 3, HH6,
WBST43	West entrance Range 3, HH6;
WBST44	Condensate pit

USDA/BARC PRCS INVENTORY
WEST SIDE/AREA B

Table 1

PRCS # West Side/8/Steam (WBST)	Location
WBST45	West entrance Range 3, HH6
WBST46	On 3rd Street near Range 3, GH6
WBST47	Cupola; at Range 2 west of 1st street and across from Bldg. 004
WBST48	Next to cupola; 3 ft x 3 ft rectangular metal cover on concrete base with lift handle; at Range 2 west of 1st. and across from Bldg. 004
WBST49	Cupola; at range 2 dr. & 1st St. and across from Bldg. 003
WBST50	Next to cupola; 3 ft x 3 ft rectangular metal cover on concrete base with lift handle;
WBST51	Cupola at Range 3 Dr. & 1st St. across from SW corner of Bldg. 002 and east of Bldg. 006
WBST52	Walk-in entrance to steam tunnel from Bldg. 001
WBST53	Cupola SW of Range 3 Dr. & 1st St. intersection; in sidewalk running east to west from 1st to 2nd streets
WBST54	Manhole with cast iron lid; by cupola in sidewalk (SW of Range 3 Dr. & 1st street intersection
WBST55	Cupola west of 1st St. and off NE corner of Bldg. 007 in grass by picnic table
WBST56	Next to cupola; manhole with cast iron lid; west of 1st St. and off NE corner of Bldg. 007 in grass by picnic table
WBST57	Cupola near front entrance to Bldg. 007 on NW side of the Bldg.
WBST58	Next to cupola manhole with cast iron lid; near front entrance to Bldg. 007 on NW side of the Bldg.
WBST59	Manhole with cast iron lid right next to Bldg. 007 front entrance; 26 diameter lid West of front steps

WEST SIDE/AREA B

Table 1

PRCS # West Side/8/Steam (WBST)	Location
WBST60	3 feet away is manhole with 22 inch diameter lid; right next to Bldg. 007 front entrance
WBST61	Manhole in sidewalk at east side of Terrace Drive and parking lot for Bldgs. 007 and 006; with 26 inch diameter lid
WBST62	A few feet away is a neighboring manhole in sidewalk; at east side of Terrace Drive and parking lot for Bldgs. 007 and 006; with 26 inch diameter lid
WBST63	Cupola in sidewalk southwest side of Bldg. 006 South parking lot
WBST64	Next to cupola in sidewalk is manhole with 26 inch cast iron lid; southwest side of Bldg. 006 South parking lot
WBST65	Manhole in sidewalk across the parking lot from SE corner of Bldg. 010A
WBST66	Neighboring manhole in sidewalk across the parking lot from SE corner of Bldg. 010A
WBST67	In sidewalk just SE of 2nd St. Range 3 Dr. intersection
WBST68	Neighboring manhole in sidewalk; just SE of 2nd St. and Range 3 Dr. intersection
WBST69	Manhole south of Range 2 Drive just east of 010A parking lot in front of (east side of Bldg)
WBST70	Manhole next to sign in front of Bldg. 008; 26 inch diameter cast iron lid
WBST71	Neighboring manhole in front of Bldg. 008; 22 inch diameter cast iron lid
WBST72	Cupola on NE corner of Bldg. 050; west of HH1
WBST73	Next to cupola is manhole with 26 inch diameter cast iron lid labeled steam; NE corner of Bldg. 050, west of HH1
WBST74	Cupola near NE entrance for Bldg. 050; HH1

USDA/BARC PRCS INVENTORY

WEST SIDE/AREA B

Table 1

PRCS # West Side/8/Steam (WBST)	Location
WBST60	3 feet away is manhole with 22 inch diameter lid; right next to Bldg. 007 front entrance
WBST61	Manhole in sidewalk at east side of Terrace Drive and parking lot for Bldgs. 007 and 006; with 26 inch diameter lid
WBST62	A few feet away is a neighboring manhole in sidewalk; at east side of Terrace Drive and parking lot for Bldgs. 007 and 006; with 26 inch diameter lid
WBST63	Cupola in sidewalk southwest side of Bldg. 006 South parking lot
WBST64	Next to cupola in sidewalk is manhole with 26 inch cast iron lid; southwest side of Bldg. 006 South parking lot
WBST65	Manhole in sidewalk across the parking lot from SE corner of Bldg. 010A
WBST66	Neighboring manhole in sidewalk across the parking lot from SE corner of Bldg. 010A
WBST67	In sidewalk just SE of 2nd St. Range 3 Dr. intersection
WBST68	Neighboring manhole in sidewalk; just SE of 2nd St. and Range 3 Dr. intersection
WBST69	Manhole south of Range 2 Drive just east of 010A parking lot in front of (east side of Bldg)
WBST70	Manhole next to sign in front of Bldg. 008; 26 inch diameter cast iron lid
WBST71	Neighboring manhole in front of Bldg. 008; 22 inch diameter cast iron lid
WBST72	Cupola on NE corner of Bldg. 050; west of HH1
WBST73	Next to cupola is manhole with 26 inch diameter cast iron lid labeled steam; NE corner of Bldg. 050, west of HH1
WBST74	Cupola near NE entrance for Bldg. 050; HH1

USDA/BARC STEAM SYSTEM PRCS INVENTORY

WEST SIDE/AREA B

Table 2

PRCS # West Side/8/Steam (WBST)	Location
WBST75	Next to cupola is manhole with 26 inch diameter cast iron lid; near NE entrance for Bldg. 050; HH1
WBST76	Cupola in front (north of) Bldg. 050, HH2
WBST77	Next to cupola is manhole with 26 inch diameter cast iron lid; in front of (north of) Bldg. 050
WBST78	Cupola in front of (north of) Bldg. 050, HH3
WBST78A	Next to cupola is manhole with 26 inch diameter cast iron lid; in front of (north of) Bldg. 050; HH3
WBST79	Cupola by Range 3, GH6, across from (east of) Bldg. 009A
WBST80	Next to cupola is manhole with cast iron lid; across from (east of) Bldg. 009A
WBST81	Cupola south side of Bldg. 014
WBST82	25 yards SE of Bldg. 011A near parking lot

EAST SIDE/BUILDING 309 BOILER PLANT

Table 2

PRCS # East Boiler Plant Steam (EBPST)	Location
EBPST1	Building 309 tunnel entrance
EBPST2	Building 308 basement entrance to tunnel
EBPST3	Building 307 basement entrance to tunnel
EBPST4	Building 306 basement entrance to tunnel
EBPST5	Rectangular manhole access to tunnel; in field by sidewalk south of Center Drive; at intersection of the 2 steam tunnels; grated metal cover over which plywood board has been placed
EBPST6	Rectangular manhole access to tunnel; in field by sidewalk south of Center Drive; a few feet north of intersection of 2 steam tunnels
EBPST7	Rectangular manhole access to tunnel; in field by sidewalk south of Center Drive; a few feet south of the intersection of the 2 steam tunnels
EBPST8	Between Bldg. 309 and 312, west of 312
EBPST9	Not in service
EBPST10	On NE corner of Bldg. 306 near intersection with Center Road and circular drive; with two 2 ft x 2 ft steel covers
EBPST11	On NE corner of Bldg. 306 near intersection with Center Road and circular drive; valve pit a few feet east of valve pit above with two 2 ft x 2 ft steel covers

EAST SIDE/DAIRY

Table 3

PRCS # East/Dairy/ Steam (EDST)	Location
EDST1	Behind (north of) Bldg. 165; concrete vault; 2 rectangular metal covers; condensate receiver pit
EDST2	Behind (north side of) Bldg. 160A; concrete and brick pit; rectangular steel cover
EDST3	Behind (north side of) Bldg. 160A; concrete vault; 2 rectangular covers
EDST4	Behind (north of) Bldg. 160C; concrete vault with 54 inch square metal cover
EDST5	In line and west of #4 above; concrete pit with 3 *x* 4 steel cover
EDST6	Behind (north of) Bldg. 162; concrete pit with 2 steel covers
EDST7	Concrete pit behind (north of) Bldg. 162; 2 steel covers
EDST8	Concrete valve pit behind (north of) Bldg. 162; steel cover
EDST9	Not in service
EDST10	Behind (north side of) Bldg. 177C
EDST11	Valve pit south side of Bldg. 177A
EDST12	In road between Bldgs. 177D and 177B; 3 x 3 metal cover
EDST13	In front of Bldg. 163A; 30 inch cast iron lid
EDST14	Across from Bldg. 163A
EDST15	On NW corner of Bldg. 179; 3 x 3 ft. metal cover

USDA/BARC STEAM SYSTEM PRCS INVENTORY

EAST SIDE/DAIRY

Table 3

PRCS # East/Dairy/ Steam (EDST)	Location
EDST16	On west side of Bldg. 173 in notch of Bldg.
EDST17	Steam pit west of Bldg. 173; between Bldg. and road
EDST18	In front of (north side of) Bldg. 167
EDST19	In front of (north of) Bldg. 168
EDST20	Not in service
EDST21	West side of Bldg. 177
EDST22	Near the SE corner of Bldg. 157
EDST23	SE corner of Bldg. 157

EAST SIDE/ANIMAL HUSBANDRY

Table 4

PRCS # East/Animal Husbandry/ Steam (EAHST}	Location
EAHST1	South side of Bldg. 204; 2 rectangular steel covers
EAHST2	East of 203D entrance on concrete base; cast iron lid

EAST SIDE/FDA

Table 5

PRCS # East/FDA/ Steam (EFDAST)	Location
EFDAST1	Not in service
EFDAST2	Not in service
EFDAST3	Not in service

EAST SIDE/POULTRY/AREA C

Table 6

PRCS # East Side/Area C Steam (ECST)	Location
ECST1	On west side (behind) Bldg. 263 in sidewalk is manhole with cast iron lid
ECST2	Steam cupola on SW corner of Bldg. 264

MISCELLANEOUS CONFINED SPACES

Water Towers, Valve Vaults and Water Meter Manholes

Tower, Vault, Manhole #	Location
288 Tower	Tower in woods overlooking BW Parkway
Bldg. 509	Water meter manhole in grass on side of Beaver Dam Road
Hayden farm tower 603	Tower in woods
Hayden farm tower	Rectangular valve vault
Hayden farm tower Bldg. 603	Rectangular water meter valve with steel cover, on one side of road
Hayden farm tower Bldg. 603	Rectangular water meter valve vault with steel cover, on opposite side of road
Bldg. 580, APHIS	Rectangular water meter vault, steel cover
Bldg. 426 tower	Across (north of) Powder Mill Road and across from Bldg. 426
Bldg. 426 tower	Rectangular valve vault
Near marijuana field	Rectangular valve vault
Center tower	Behind Bldg. 308 (may be numbered 305)
Center tower	Rectangular valve vault
Secret Service tower	300,000 gallon capacity; No Access
Secret Service tower	Rectangular valve vault; No Access
APU tower	Not in use #1106
APU tower	Rectangular valve vault

USDA/BARC PRCS INVENTORY

Water Treatment Plant, 310 East Side

Tower, Vault, Manhole #	Location
Water tank	40,000 gallon water tank
Water tank	Access manhole
Water tank	Access manhole
Valve vault	2 ft x 2 ft steel plate access cover

Abandoned Water Storage Reservoir at Building 200/East Side

Valve Vault & Manhole #	Location
Valve vault	South of Bldg. 200: 3 ft x 3 ft steel access covers
Valve vault	South of Bldg. 200: 3 ft x 3 ft steel access covers
Valve vault	South of Bldg. 200: 3 ft x 3 ft steel access covers
Manhole	40 ft west of valve vaults

Water Meter Manholes and Vaults/East Side

Valve Vault & Manhole #	Location
Valve shut-off	Not in service
Valve shut-off	Not in service
Water meter	Not in service

USDA/BARC PRCS INVENTORY

Water Meter Manholes and Vaults/West Side

Valve Vault & Manhole #	Location
Meter manhole	Library; 24 inch diameter cast iron lid sprayed blue; at the intersection or RT. 1 and entrance to library; across from the entrance drive to the ARS by Bldg. 001; water meter main and valve off

Boiler Survey Inventory of Confined Spaces

This report is a summary of the boiler survey which involved interviewing boiler plant operators characterizing the boilers at the USDA/BARC facility in Beltsville, Maryland.

1.0 Background

There are six steam plants at the USDA-ARS facility at the following locations:

Building 309
Building 014
Building 261
Building 165

Boilers were not inspected at each of the above locations because access to some of the buildings was not possible at the time of the survey. There are also boilers in other buildings, including, but not limited to, Building 254. The boiler in Building 254 is about 5 ft x 4 ft x 5 ft. After unscrewing the metal panels, work can be done without entering. This boiler is not considered a confined space.

2.0 Boilers

a. Building 014 Boilers (PRCSs)

The boiler plant operator at Building 014 was interviewed regarding the operation of the boilers at that location. There are five fire-tube boilers at this facility in which water circulates around the tubes. There is a 15-inch diameter inspection plate at the top of the boiler.
Entry via this inspection hatch would only be done if retubing the boiler, for example, having someone go in to hold the tube and help guide it into place. Such entry would be done by a contractor. These boilers

are permit-required confined spaces. They are not labeled because the surfaces become too hot and the label will be destroyed. The boilers cannot easily be accessed by unauthorized persons.

b. Building 309 Boilers (PRCSs)

The boiler plant operator at Building 309 was interviewed regarding the operation of the boilers at that location. Boilers were inspected in Building 309. There are four fire- tube boilers. No entry would be done on the water side of the boiler. Entry would be done to inspect and/or resurface the refractory. The four boilers can each be opened up by removing a number of large bolts on the rear cover. This cover is the same dimensions as the entire width of the boiler allowing for the entire back portion of the boiler to swing open on a hinge - approximately 15 feet - and then tied off. Workers must bend over somewhat to enter the boiler. These boilers are permit-required confined spaces. They are not labeled because the surfaces become too hot and the label will be destroyed. The boiler cannot be easily accessed by unauthorized persons and because a number of large bolts must be removed.

c. Building 165 Boilers (PRCSs)

There are 3 boilers in this building. The covers can be unbolted at either end, allowing workers to go from one end of the boiler to the other. Entry normally is not done. Typically the boilers are cleaned (dusted) from outside by using a garden hose. These boilers are permit-required confined spaces. They are not labeled because the surfaces become too hot and the label will be destroyed. The boilers cannot be easily accessed by unauthorized persons and because a number of large bolts must be removed.

d. Building 261 (PRCSs)

There are 2 boilers inside this building. The boilers are about 10 feet long and the covers are about 4 feet in diameter. These boilers are

permit-required confined spaces. They are not labeled because the surfaces become too hot and the label will be destroyed. The boilers cannot be easily accessed by unauthorized persons and because a number of large bolts must be removed.

e. Buildings 085 and 320A

The boilers at these locations are assumed to be permit-required confined spaces. The boilers should not be labeled because the surface becomes too hot and the label will be destroyed. The boilers cannot easily be accessed by unauthorized persons.

3.1 Hazards

The principle hazards include:

a. Potential for reduced oxygen and/or toxic atmosphere

b. Potential for exposure to hazardous energy sources including electrical, mechanical, hydraulic, scalding steam

c. Engulfment

4.1 Recommendations

a. Work in these spaces involves confined space entry and appropriate steps must be taken to comply with the OSHA Permit-Required Confined Space Standard (1910.146), including, but not limited to:

1) Entry permit

2) Air monitoring for LEL, oxygen, carbon monoxide, and hydrogen sulfide

3) Ventilation of the space

4) Emergency procedures and rescue

5) A minimum of two people for each entry (entrant and attendant/entry supervisor)

b. The following is a standard operating procedure for entering the boilers:

1) Notify appropriate persons including your supervisor and fill out a confined space entry permit. Ensure a minimum of two people (entrant and attendant/entry supervisor) are present during the entire procedure.

2) Turn off all power, gas, oil feeds and steam connections to the boiler and lock them out.

3) Unbolt the rear cover or access hatch and remove it. Test inside the boiler for combustible gases and vapors, carbon monoxide and oxygen levels.

4) Use a portable blower (1000 CFM) and ventilate the inside of the boiler for a minimum of 10 minutes.

5) Test the air in the boiler again. Continue to ventilate.

6) A retrieval system is not necessary since the boilers are 6 to 8 feet in width and 10 to 15 feet in depth.

7) Enter the boiler and complete the work. Use low-voltage lighting, flashlights or battery-powered direct beam lights to illuminate the boiler work area. An attendant must be present and the boiler must be continuously ventilated.

5.1 Other Permit-Required Confined Spaces at Building 309

Because the boilers at Building 309 were originally burned coal, there are several associated tanks or structures in this area. Although these structures are never used or entered, they are permit-required confined spaces. These structures are located above the boilers in Building 309 near the ceiling and consist of:

a. A huge silver-gray tank in the shape of an inverted cone: After cleaning the boiler out, the ashes were exhausted into this tank.

b. Another huge silver-grey rectangular bin used for the storage of coal.

c. A huge black rectangular bin used for breaching - the stacks on the coal boilers were piped into this structure.

Appendix F

BARC Research Farm Services and Research Animal Services

USDA/BARC PRCS INVENTORY
DAIRY MANURE PROCESSING
AND
WASTE TREATMENT SYSTEM

Table 1

PRCS #	Location
Manure Pit 1*	Building 178-2; 0%LEL,20.8% 02, 0 PPM CO, 0 PPM H2S
Manure Pit 2*	Building 178-2; 0%LEL, 20.8% 02, 0 PPM CO, 0 PPM H2S
Manure Pit*	Lot 1; 0%LEL, 20.8% 02, 0 PPM CO, 0 PPM H2S
Liquid Waste Pit*	Building 184; 0%LEL, 20.8% 02, 0 PPM CO, 0 PPM H2S
Anaerobic Digester	**north side of Building 184 Not in Service**
Open-top storage tank	north side of anaerobic digester
Solids Storage bin	south side of Building 184: assumed to be PRCS

Dairy Milk Bulk Storage Tank Survey
Inventory of Confined Spaces

1.0 Background

The milk storage tank, located in Building 178-1is a permit-required confined space. The bulk storage tank is a 4000 gallon stainless steel tank with an 18-inch diameter opening at the top of the tank. The tank can be entered, and USDA/BARC employees have on occasion entered the tank to clean the tank by hand when the automatic washer has malfunctioned.

A low RPM agitator is located at the bottom of the tank. The agitator is unguarded. Usually entry is not required during cleaning. When the milkman comes to clean the tank, he hooks up an automatic washer which contains detergent, wash, and acid rinse

2.0 Hazards

The principal hazards in this space consist of:

a. Electrical shock hazard

b. Hazardous mechanical energy (the unguarded agitator)

c. Potential for reduced oxygen

d. Potential for exposure to chemicals in the detergent and acid rinse

e. Fall hazards during entry, while working in the space, and exiting

3.0 Recommendations

a. The milk bulk storage tank meets the OSHA definition of a permit-required confined space (PRCS). However, entry can be made using the alternative procedures specified in 1910.146(c)(5) as long as the only hazard or potential hazard_ is that of a hazardous atmosphere, all hazardous energy sources have been isolated and locked out prior to entry, and there are no other non-atmospheric hazards.

b. Appropriate steps must be taken to comply with the alternative procedures outlined in the OSHA standard 1910.146(c)(5) including but not limited to:

1) Continuously monitoring the space with a calibrated direct reading meter;

2) Continuous ventilation of the space before and during entry;

3) Entrant has received permit-required confined space entrant training;

4) A minimum of two people for the entry (the USDA/BARC policy requires two people for entries done under alternative procedures).

c. The OSHA Lockout/Tagout standard (1910.147) must be complied with prior to entering the tank. This standard requires the control of hazardous energy when employees conduct servicing or maintenance of equipment. Lockout of the energy isolating device controlling the transmission energy to the agitator must be done prior to entry. Controlling residual (stored) energy is not applicable because the agitator is a low RPM agitator. A guard or barrier should be placed over the agitator to prevent slipping or tripping on this mechanism. Such entry does require continuous monitoring with a calibrated, direct-reading instrument and continuous forced air ventilation.

d. Work inside the bulk storage tank may require the use of personal protective equipment such as electrical protective equipment, eye and face protection, respiratory protection, head protection, and foot protection and must meet the following OSHA standards:

1) 1910.132 - General PPE

2) 1910.133 - Eye and Face Protection

3) 1910.134 - Respiratory Protection

4) 1910.135 - Head Protection

5) 1910.136 - Foot Protection

USDA/BARC PRCS INVENTORY
FEED BINS/TANKS

Table 1

Feed Bin # (FB)	Feed Bin/Tank	Location
FB47	Feed Tank at Bldg 178-2	Dairy
FB48	Feed Tank at Bldg 178-2	Dairy
FB49	Feed Tank at Bldg 178-2	Dairy
FB50	Feed Tank at Bldg 178-2	Dairy
FB51	Feed Tank at Bldg 203 East Wing Fiberdome	Swine
FB52	Feed Tank at Bldg 203 West Wing Fiberdome	Swine
FB53	Feed Tank at Bldg 203D Fiberdome	Boar Barn
FB54	Feed Tank at Bldg 208 Fiberdome	Farrowing Barn
FB55	Feed Tank at Bldg 208 Fiberdome	Swine
FB56	Feed Tank at Area 500 Fiberdome	Swine **Not in Service**
FB57	Feed Tank at Bldg 229 Metal (101)	Poultry
FB58	Feed Tank at Bldg 229 Metal (102)	Poultry

USDA/BARC/RSS
FEED AND FORAGE UNIT (FFU)/PESTICIDES
INVENTORY OF CONFINED SPACES

1.0 Background

a. Tanker trucks{PRCSs)

The Feed and Forage Unit operates and maintains {4) 6000gal tanker trailer/ tanker trucks to haul liquid manure from the Dairy lagoon to crop fields and Waste Water Sludge from the water treatment plant to the Sludge field each year. Tankers are approximately 30' long by 6 foot in diameter. There are access hatches in the tops of the tanks. Entry has never been necessary all workings are external so there is no need to enter the tank. Clogs are dealt with from outside of the tank with air injection. These tankers are permit-required confined spaces.

b. Manure Spreader

The Liquid Manure spreader is a PTO-driven self-agitating 5300 gallon spreader pulled by a large tractor that applies the liquid manures to the crop fields and the wastewater sludge to the sludge field each year. The tank has never been entered; all of the maintenance components are external so entry is not necessary. This is a permit-required confined space.

c. Building 182 Feed Mill

Building 182 is a working industrial size feed mill used to mix, pellet, store and bag feed products for the research center. The building has 24 upright bulk feed ingredient storage tanks that all have roof top access. The bins range in size from 8-44 ton holding capacity. There is also a 5000 bushel dry grain storage bin located outside the building. All of the bins have access for entry for repairs or cleaning. Entry has only been made by contractors licensed to do so.

d. Trucks for use with fertilizer machines (PRCSs)

Two trucks are used in conjunction with the fertilizer machines in field. One truck uses an 1100 gallon plastic tank with an access hatch at top. The stainless-steel tanker truck was formerly used to haul milk, and as an access hatch at the top. These tanks have never been entered, but it is possible to enter them. These tanks are permit-required confined spaces.

2.0 Hazards

The principal hazards in these spaces consist of the following:

a. Potential for exposure to hazardous chemicals (pesticides and/or fertilizer) through inhalation and skin contact

b. Potential reduced oxygen

c. Potential for explosive methane gas

d. Potential for carbon monoxide

e. Potential for hydrogen sulfide

f. Engulfment

g. Fall hazards during entry, work, and exit

3.0 Recommendations

a. Work in these spaces involves confined space entry and appropriate steps must be taken to comply with the OSHA Permit-Required Confined Space Standard (1910.146), including, but not limited to:

    1) Entry permit;

    2) Air monitoring for LEL, oxygen, carbon monoxide, and hydrogen sulfide;

    3) Ventilation of the space;

    4) Emergency procedures and rescue;

    5) Fall protection and retrieval for those tanks greater than 5 feet in depth

    6) Minimum of two people for entry (entrant and attendant/entry supervisor)

b. Work in these confined spaces must require the use of personal protective equipment such as appropriate respiratory protection (pressure-demand airline respirators with escape/self-contained breathing apparatus), chemical protective clothing, eye and face protection, protective gloves, and head protection. This personal protective equipment (PPE) must meet the following OSHA requirements:

    1) 1910.132 General PPE requirements including selection, use and training

2) 1910.133 Eye and Face Protection

3) 1910.134 Respiratory Protection

4) 1910.135 Head Protection

5) 1910.136 Foot Protection

c. Because pesticides and fertilizer are extremely toxic via inhalation and skin contact, entry into the tanks should not be done. Farm operations personnel have never entered any of the pesticide or fertilizer tanks and trucks.

USDA/BARC/RSS
INCINERATOR SURVEY
INVENTORY OF CONFINED SPACES

1.0 Background

There are incinerators at the USDA/BARC facility at the following locations:

Multi-Species Unit (MSU)/Building 204

Animal Parasitology Unit/Building 1320 – New Unit Installed

All the incinerators meet the definition of a permit-required confined space.

a. Animal Parasitology Unit (PRCS)

The incinerator chamber at the Animal Parasitology Unit (Building 1320) is approximately 4.5 feet wide and 6 feet deep. To clean the incinerator, the door is opened and a long-handled shovel or hoe is used while standing in front of the incinerator. Entry would only be done by an outside contractor in order to redo the lining on the interior of the incinerator.

b. Building 204 (PRCS)

The incinerator at Building 204 is a permit-required confined space. It is rated at 1000/pounds per hour and is loaded from the top. The incinerator is fueled by natural gas. Normally entry is not done. To clean the incinerator, the door is opened and a long-handled shovel or hoe is used while standing in front of the incinerator. This door is 5 feet by 4 feet. The incinerator is about 10 feet long. Entry would only be done by an outside contractor in order to redo the lining on the interior of the incinerator.

2.0 Hazards

The principle hazards include:

a. Potential for reduced oxygen and toxic atmosphere (carbon monoxide)

b. Potential for fire and explosion

c. Potential for exposure to hazardous energy sources including electrical, mechanical, hydraulic, etc.

d. Engulfment

3.0 Recommendations

a. Work in these spaces involves confined space entry and appropriate steps must be taken to comply with the OSHA Permit-Required Confined Space Standard (1910.146), including, but not limited to:

1) Entry permit;

2) Air monitoring for LEL, oxygen, carbon monoxide, and hydrogen sulfide;

3) Ventilation of the space;

4) Emergency procedures and rescue;

5) A minimum of two people for each entry (entrant and attendant/supervisor)

b. The following is a recommended standard operating procedure for working inside the incinerators:

1) Notify appropriate persons including your supervisor and fill out a confined space entry permit. Ensure a minimum of two people (entrant and attendant/entry supervisor) are present during the entire procedure.

2) Turn off and lock out all hazardous energy sources to the incinerator: all power, gas, oil feeds, hydraulic, etc.

3) Unbolt the cover and open the door. Test inside the incinerator for combustible gases and vapors, carbon monoxide and oxygen levels.

4) Use a portable blower (1000 CFM) and ventilate the inside of the incinerator for a minimum of 10 minutes.

5) Test the air inside the incinerator again. Continue to ventilate.

6) A retrieval system is not necessary since the incinerators are 6 to 8 feet in width and 10 to 15 feet in depth.

7) Enter the incinerator and complete the work. An attendant must be present and the incinerator must be continuously ventilated.

Appendix F

BARC

Information Technology

USDA/BARC INVENTORY WESTSIDE/AREA B TELECOMUNICATIONS CABLE RUN ALTERNATIVE PROCEDURES

Table 1

Manhole #	Location
GMH32	In the intersection of 1st Street and Range 2 Drive; big vault;
GMH31	On 1st Street west of Bldg. 003
RT	Proximal to west side of Bldg. 002
GMH39	In 1st St., but just west of 12A
GMH12A	Near 1st Street, on southwest side of Bldg. 002
GMH13	South of Bldg. 006 and west of 1st Street; on SE corner of parking lot
GMH14	West of 13 near parking lot for Bldg. 006
GMH25	Proximal to east side of parking lot for Bldg. 007
GMH26	Proximal to north side of Bldg. 007
GMH15	In grass just west of Range 3 Drive and 2nd St. intersection
GMH15A	South of 15 and proximal to east side of GH1
GMH23	Near northeast corner of Bldg. 008
GMH27	Near northwest corner of Bldg. 050
GMH16	Just south of Range 3 Drive between 2nd and 3rd Street
GMH17	In grass just southwest of Range 3 Drive and 3rd St. intersection
GMH18	Just south of Range 3 Drive and 4th St. intersection near substation
GMH18A	South of Range 3 Drive and west of 18 across from Bldg. 014
GMH19	Sn grass south of 3rd St. and between east side of Bldgs. 048 & 049

USDA/BARC INVENTORY WESTSIDE/AREA B TELECOMUNICATIONS CABLE RUN

Table 1

Manhole #	Location
GMH19	In grass south of 3rd St. and between east side of Bldgs. 047A & 047
GMH20	In grass south of 3rd St. and between Bldgs. 043 and 046
GMH21	In grass south of 3rd St. and on east side of Bldg. 046A
GMH33	In Range 2 Drive and 2nd St. intersection across from parking lot behind Bldg. 010A
GMH34	In Range 2 Drive and 3rd Street intersection
GMH35	In Range 2 Drive across street from northwest corner of Bldg. 012
GMH36	In 2nd Street just west of (behind) Bldg. 010A
GMH28	In grass on north side of North Drive across street from HH3
GMH29	In grass on north side of North Drive across street from HH11
GMH30	In grass just south of par-king lot for Bldg. 011A

USDA/BARC INVENTORY WESTSIDE/AREA B TELECOMUNICATIONS CABLE RUN ALTERNATIVE PROCEDURES

Table 1

Manhole #	Location
E309TC	
1	Just east of Bldg. 306 near front entrance and just behind Bldg. Sign
2	Just north of intersection of two steam tunnels in grass east of Center Road
3	On east side of Bldg. 308 near front entrance
4	SW corner of Bldg. 309

Appendix F

UNSA

Facilities Unit

USDA/NATIONAL ARBORETUM
PRCS INVENTORY

Table 1

National Arboretum PRCS	Location
1- Pump filtration vault for reflecting pool*	**Removed due to upgrade to vault**
2 - Water meter manhole (owned by District)*	near New York Avenue; Behind Bldg. 012; 30 inch cast iron lid; contains valve off; 10 feet deep and filled with water; 0% LEL, 20.8% 02, 0 PPM CO, 0 PPM H25
3 - Water meter manhole (owned by District)*	near New York Avenue behind Bldg. 012 and about 5 feet away from manhole noted above; 8- 10 feet deep and filled with water; 22 inch cast iron lid; 0% LEL, 20.8% 02, 0 PPM CO, 0 PPM H2S
4 - Water meter manhole (owned by District)	R Street
5 - Water meter manhole owned by District)	R Street
6 - Well Vault #1*	East of Bldg. 012; Concrete vault 3. 5 ft x 3.5 ft and 10 feet deep with built in ladder; 2.5 x 2.5 ft steel lid; contains valve shut-off; 4% LEL, 20.8% 02, 0 PM CO, 0 PPM H25
7 - Well Vault #2	
8 - Well Vault #3	

USDA/NATIONAL ARBORETUM
ALTERNATIVE PROCEDURES INVENTORY

Table 2

Catch basins all have brick construction unless otherwise noted.

National Arboretum Catch Basins NACB#	Location
1	Catch basin at capital columns; with concrete lid; discharges into woods across the road;
2	Catch basin; corner of Valley & Meadow by stop sign; concrete top with manhole cover;
3	Catch basin north of Springhouse Road & Hickey
4	Catch basin south of Springhouse Road & Hickey
5	Catch basin on Hickey near pond; concrete structure with concrete lid with lift handles
6	Catch basin across (southeast) of the above area on Hickey
7	Manhole with cast iron lid; on north side of Conifer Road between Springhouse and Dogwood Circle
8	Near above manhole on Conifer Road
9	Catch basin at Dogwood Circle
10	Catch basin at Dogwood Circle
11	Catch basin by parking area on Hickey Road east of Gotelli Dwarf Conifers
12	Catch basin on Hickey hill.
13	Catch basin on Hickey Hill
14	Catch basin near Asian Parking Lot
15	Catch basin near Asian Parking Lot

USDA/NATIONAL ARBORETUM
ALTERNATIVE PROCEDURES INVENTORY

Table 2

Catch basins all have brick construction unless otherwise noted.

National Arboretum Catch Basins NACB#	Location
16	Manhole on Hickey Hill south of Asian Parking lot on east side of road; near wooded area
17	Catch basin on Hickey Hill south of Asian Parking lot on west side of road
18	Catch basin near Hickey Hill & Holly Spring
19	Catch basin near Hickey Hill & Holly Spring
20	Catch basin near Maryland Avenue NE entrance in wooded area
21	**Removed because entrance close and catch basin removed**
22	Catch basin across road from Security Bldg. by Maryland Avenue NE entrance
23	Catch basin across from Security Bldg. by Maryland Avenue NE entrance; but in between the 2 roads
24	Catch basin on south side of Rhododendron Valley Road just past little pond
25	Catch basin on east side of Azalea Road before Viburnum
26	Catch basin on east side of Azalea Road after Viburnum
27	Catch basin on Azalea Road at crest of hill heading north
28	Catch basin on east side of Azalea
29	Catch basin on east side of Azalea just south of parking lot
30	Catch basin at Azalea & Mt. Hamilton Road intersection

ALTERNATIVE PROCEDURES INVENTORY

Table 2

Catch basins all have brick construction unless otherwise noted.

National Arboretum Catch Basins NACB#	Location
31	Catch basin on Mt. Hamilton Road near drive going to Lee Azalea Garden
32	Catch basin on Mt. Hamilton Road with cast iron lid
33	Catch basin on Mt. Hamilton Road with cast iron lid just before circular drive
34	Catch basin at end of Mt. Hamilton Road
35	Catch basin on Eagle Nest Road by the Lee Azalea Garden
36	Catch basin on Eagle Nest Road by the Morrison Azalea Garden
37	Catch basin on Eagle Nest Road by the Morrison Azalea Garden
38	Catch basin at Eagle Nest and Azalea
39	Catch basin on Valley Road just north of Parking area
40	Catch basin on Valley near Hickey Run; cast iron lid
41	Catch basin on Holly Spring south of Hickey Lane
42	Catch basin on Holly Spring midway between Meadow and Hickey Hill Road
43	Catch basin on Holly Spring midway between Meadow and Hickey Hill Road; cast iron lid
44	Catch basin on Holly Spring near parking area
45	Catch basin on Holly Spring near Parking area

ALTERNATIVE PROCEDURES INVENTORY

Table 2

Catch basins all have brick construction unless otherwise noted.

National Arboretum Catch Basins NACB#	Location
46	Catch basin at Kingman Lake overlook by Crabapples
47	Catch basin at Kingman Lake overlook by Crabapples
48	Catch basin in parking lot across from Administration Bldg
49	Catch basin in parking lot past Administration Bldg
50	Catch basin in parking lot past Administration Bldg
51	Catch basin in parking lot past Administration Bldg
52	Catch basin by Bldg 012
53	Catch basin by Bldg 012
54	Catch basin with cast iron lid by Bldg 012
55	Trough drain; 5 ft deep near kilns
56	Above ground 4 feet high concrete catch basin with blue metal grate in field by parking lot for kiln area
57	Catch basin by New York Avenue entrance
58	Catch basin behind Administration Bldg